

METHODOLOGY

GS4GG PAA M400-XX

SDG 13

SUSTAINABLE TRANSFORMATION THROUGH AFFORESTATION, REFORESTATION AND REVEGETATION (STARR)

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SUMMARY

This methodology provides a framework for project activities that remove and store carbon dioxide (CO₂) by establishing woody vegetation - including trees, shrubs, palms, and bamboo—on eligible non-forest land. The STARR methodology applies to Afforestation, Reforestation, and Revegetation (ARR) activities. These activities can be implemented as standalone projects or as part of broader land-use changes such as agroforestry and silvopastoral systems.

The methodology requires quantifying carbon through direct measurements of biogenic carbon pools. These pools include aboveground and belowground biomass, and optional dead organic matter. The methodology also covers the monitoring of Soil Organic Carbon (SOC) sequestered as a result of project activities.

The methodology is designed to align with the Paris Agreement and the Gold Standard for the Global Goals (GS4GG) requirements. It uses Downward Adjustment Factors (DAF) to update baselines over time and support the climate goals of the Host Country. This methodology shall follow GS4GG Land Use & Forests (LUF) Activity Requirements and safeguards to ensure permanence, avoid double counting, and manage leakage risks.

ACKNOWLEDGEMENTS

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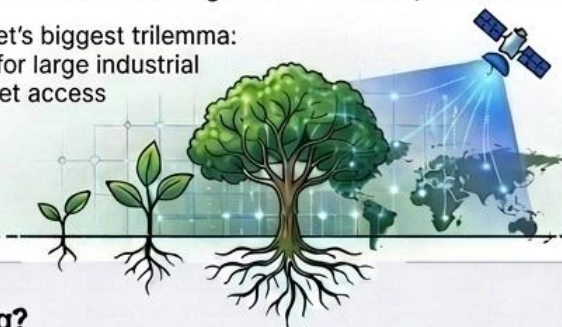
Gold Standard would also like to thank our expert panel for their valuable feedback during the review process.

Public Consultation

STARR: The Next Generation of Carbon Removals

Sustainable Transformation Through Afforestation, Reforestation & Revegetation

"Solving the carbon market's biggest trilemma: delivering scientific rigor for large industrial projects + equitable market access for smallholder farmers in the Global South."



What are you planting?

Modular Architecture



Commercial Forestry



Restorative Forestry



Production Agroforestry



Subsistence Agroforestry



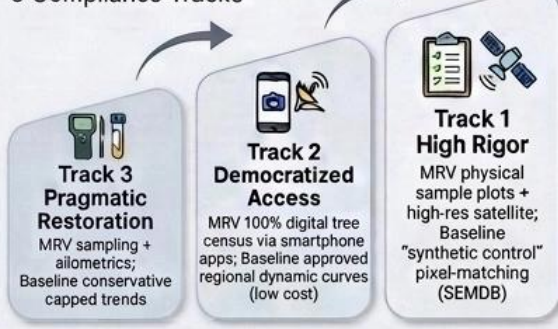
Revegetation



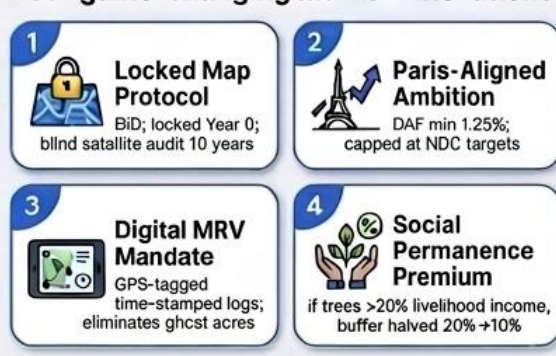
+ Sub-Module
A Smallholder
Aggregation

How do you measure it?

3 Compliance Tracks



Four game-changing market innovations



Net Carbon Math equation

Gross Biomass Growth — Activity Emissions — Adjusted Baseline — Leakage
= Net Issuable GSVERs (minus 10–20% buffer)

Credits only for pure removals (no avoided degradation)

Eligibility & Safeguards



Eligible: non-forest land >10 years



Food security: yield maintenance test



Excluded: wetlands, peat drainage, invasive species, conversion of native ecosystems

Gold Standard for the Global Goals | Document: GS4GG PAA M400-XX

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1 | KEY INFORMATION

1.1 | Key Information Table

1.1.1.1 | The following table describes the key information for the application of methodology.

Table 1. Key information

Term	Description
Activity ¹	Activities involve the planting, and / or assisted natural regeneration of trees, shrubs, palms and bamboo on land that has been non-forest for at least 10 years prior to the start date. The activity sequesters carbon in biomass and soil pools while complying with the specific rules of its assigned Compliance Track.
Type of GHG mitigation measure(s)	☒ Nature based carbon dioxide removal
Mitigation type	☒ Emission removals
Applicable activity strategies	These are the key activity types which can be selected based on the design of an activity. These are detailed in the methodology. This includes afforestation, reforestation and revegetation. Avoiding emissions from reducing degradation or deforestation are not eligible in this methodology.
Applicable activity scale	☒ Micro scale (e.g., ≤ 10,000 tCO ₂ e per year and not more than 500 ha) ☒ Small scale (e.g., ≤ 60,000 tCO ₂ e per year and not more than 10,000 ha) ☒ Large scale (e.g., > 60,000 tCO ₂ e per year or more than 10,000 ha)
Sectoral Scope	14. Forestry
Activity Requirement	Land use and forestry activity requirement
Activity start date	The earliest date on which the first trees are planted or the first activity associated with assisted natural regeneration began on the field, within the activity area. In cases where both plantation and ANR are part of the activity, the earliest date shall be considered.

¹ The terms 'Activity', 'Project' and 'Project Activity' refer to the project certified by GS4GG and are used interchangeably

Crediting Period start date	The Crediting Period start date is either the Activity start date, or three years prior to the date of Design Certification, whichever occurs later.
Crediting period length	40-100 years
Geographical applicability	Global
Limitations	NA

1.2 | Strategy and Compliance Track Selection Matrix

- 1.2.1. | To apply this methodology, the Activity Developer shall identify the strategy and scale based on the ARR activity design. With these selections, the appropriate Module and associated Compliance Track is identified from the Track Selection Matrix.
- 1.2.2. | **Debundling:** If Track 2 or 3 is selected, the Activity Developer shall demonstrate that the project is not debundled from a large-scale project.
- 1.2.3. | The six possible strategies are explained below:
- 1.2.3.1 | **Commercial Forestry:** This strategy encompasses afforestation or reforestation implemented for the commercial production of timber, pulp and fuel wood. Importantly the strategy shall not include an integrated agricultural component.
- 1.2.3.2 | **Restorative Forestry:** This strategy aims to reestablish forests and woodlands in areas that were once forest but will not meet the definition of forest for at least 10 years before the start date. Design may include the planting of saplings, assisted natural regeneration and natural successional processes and may include developing biodiverse corridors. Importantly, no commercial extraction of forest products is permitted.
- 1.2.3.3 | **Production Agroforestry:** Integrated land use systems including agroforestry or silvopastures are examples of this strategy. The forestry component is designed for commercial production.
- 1.2.3.4 | **Subsistence Agroforestry:** Multi-species integrated land use systems (e.g., agroforestry or silvopastoral) primarily for local food security, fuelwood, and soil health with minimal external commercial sales.
- 1.2.3.5 | **Revegetation:** The distinctive feature of this strategy is the use of both tree and non-tree species (e.g. bamboo, or shrubs) where the eligible area after implementation does not meet the forest definition.
- 1.2.3.6 | **Smallholder Aggregation:** Where the ARR activities are implemented within lands where smallholders have a legal right. This also includes activities in communal lands, village forests etc. In such cases where land is not owned by smallholders, a community-led institution (such as a farmer cooperative) shall be involved.

1.2.4.| These strategies are arranged into 5 Modules and 1 Sub-module as summarised in Table 2 below.

Table 2. Modules and Sub-Modules based on the strategy

Module	Strategy	Primary Objective
Module 1 (M1)	Commercial forestry	Timber & Commercial production
Module 2 (M2)	Restorative Forestry	Restoration of forests and woodlands
Module 3 (M3)	Production Agroforestry	Integrated land use with commercial production
Module 4 (M4)	Subsistence Agroforestry	Integrated land use systems for subsistence purposes
Module 5 (M5)	Revegetation	Trees in small clusters
Sub-Module A	Smallholder Aggregation	Livelihood support

1.2.5.| An activity design shall apply a module based on the strategy. Exceptions are:

- Module 5 (revegetation) is allowed to be stacked with Modules 1, 2, 3 or 4.
- Activities under all modules, except Module 1, may be developed as standalone smallholder aggregation under the provisions of sub-module A of this methodology.

Table 3. Modules and Sub-Modules Stacking Applicability

Module	+ Module 5 (M5)	+ Sub-Module A: Smallholder Aggregation
Module 1 (M1)	Allowed in arid/semi-arid zones where forest definition may not be met but biomass is harvested. Only Track 1 allowed.	Not Allowed
Module 2 (M2)	Allowed , including for restoring degraded shrublands or savannas.	Allowed: Smallholder led restoration of communal forests or degraded lands.
Module 3 (M3)	Allowed. Examples include using trees, shrubs/bushes for windbreaks or shelterbelts in commercial orchards.	Allowed: The model where small farms feed into a central commercial supply chain through farmer cooperatives or similar institutional mechanisms.
Module 4 (M4)	Allowed: Examples include home-gardens incorporating tree and non-tree woody species for fodder/fuelwood.	Allowed.

- 1.2.6.| Three compliance monitoring Tracks (Tracks) are allowed based on the Module applied and the scale of the activity:
- 1.2.7.| Track 1: Dynamic baseline estimated using synthetic control (SEMDB).
- 1.2.8.| Track 2: Regional dynamic aggregates.
- 1.2.9.| Track 3: Conservatively capped static exception.
- 1.2.10.| Tracks allowed for each module are detailed in Table 4 below. If Module 5 (Revegetation) is stacked, the track allowed for the other module with which it is stacked (i.e., Module 1, 2, 3 or 4) will be applicable to the Revegetation component also.

Table 4. Track Permitted per Module and Scale

Module	Strategy	Large scale	Small scale	Small-holder
M1	Commercial forestry	Track 1	Track 1	N/A
M2	Restorative forestry	Track 1 / 2	Track 1 / 2	Track 1 / 2 / 3
M3	Production agroforestry	Track 1	Track 1 / 2	Track 1 / 2 / 3
M4	Subsistence agroforestry	Track 1 / 2	Track 1 / 2	Track 1 / 2 / 3
M5	Revegetation (with no stacking)	Track 1	Track 1 / 2	Track 1 / 2 / 3

Questions for Stakeholders:**Track 2: Regional Dynamic Aggregate**

Concept: To enhance inclusivity for small scale and restorative or subsistence projects lacking GIS-matching capabilities, this methodology proposes a "Track 2" option. Under this track, project developers may "subscribe" to a Regional Dynamic Curve approved by the Gold Standard or a recognized national authority for a specific Eco-Zone and Land Class (e.g., "Degraded Miombo Savanna").

Aggregated NFI Fallback: In the absence of a published curve, developers may derive sequestration rates from National Forest Inventory (NFI) data.

Requirement: Data must be aggregated from a minimum of 30 plots within the same administrative district or zone and must have been updated within the last five years.

An **illustrative** application of Track 2 across modules is provided below:

1. Modules 1 and 2: Utilization of regional average rates for non-project conversion (non-forest to forest) and associated carbon stock changes.
2. Modules 3 and 4: Utilization of regional average carbon stock changes on non-forest lands.
3. Module 5: Utilization of regional average carbon stock changes in forests, specifically excluding data from natural parks or protected areas to ensure additionality and representativeness.

Stakeholders are invited to provide feedback on the viability of the Track 2 approach. We specifically seek input on data availability and welcome proposals for alternative data sources that meet the required rigor for Track 2.

Question 1 Data Integrity & Conservativeness: To account for the inherent uncertainty in regional averages compared to project-specific GIS matching, should a conservativeness factor (e.g., a 10-20% discount) be applied to the Regional Dynamic Curve? If so, what level of discounting is appropriate?

Question 2 NFI Fallback Criteria: Is the requirement of 30 plots within a 5-year window realistic for most small or micro-scale project locations? If these criteria cannot be met, what alternative statistical thresholds or 'proxy' data sources (e.g., IPCC Tier 2 defaults) should be permitted?

Question 3 Regarding Module 5, does the exclusion of protected areas and national parks sufficiently mitigate the risk of overestimating sequestration rates in degraded landscapes, or should further land-use history filters be applied?

Question 4 Eligibility thresholds: Should Track 2 be strictly limited to Small/Micro-scale projects, or should it be available to larger projects for Module 3 and Module 5 also, in 'Data-Poor' regions where GIS-matching data is objectively unavailable?

Question 5 Dynamic Updating: How frequently should a 'published curve' be re-validated to remain 'Dynamic'? Should the methodology mandate an update every 5 years, or should it be linked to the baseline renewal (10 years)?

2| DEFINITIONS

2.1.1.1| The definitions outlined in the [Glossary of Gold Standard for the Global Goals](#) and the [Forestry Activity Requirement](#) shall apply, in addition to those outlined below:

Table 5. Terms and definitions

TERM	DEFINITION
Activity	An action or combination of actions by the activity developer or project participants that remove and store more carbon dioxide (CO ₂) in woody vegetation and soil organic carbon on non-forest land. Activities may include planting, and / or assisted natural regeneration of trees, shrubs, palms and bamboo.
Activity Boundary	The geographical delineation of the Project Area, in which the activities occur. The activity boundary covers the full lifecycle of the removals, including land preparation, planting, management, and the durable storage of carbon in living and dead biomass and soil pools.
Activity Strategy	The classification of the project activity into one of the distinct strategies (Commercial forestry, Restorative forestry, Production/Subsistence agroforestry, Revegetation).
Baseline Validity Period	The fixed period of 10 years during which the baseline removals are calculated based on parameters set at the start of the period.
Commercial plantation	A plantation established to generate forest products, for example, timber, paper and pulp, or palm oil, for sale into commercial markets. All palm oil plantations are considered to be commercial plantations. If timber harvesting is not systematic or planned, but for use by the smallholders, it is deemed not a commercial activity. Non-timber forest products harvested in a sustainable and non-destructive manner, limited to fruits or flowers, for self-consumption or sale by smallholders is not deemed as a commercial use.
Compliance Track	The specific set of methodological processes (Baseline approach, Additionality assessment, Monitoring method, and evaluation of Safeguards) assigned to an activity \based on its strategy and scale.
Digital MRV System	A technology platform (e.g., mobile app and cloud database) used to record, aggregate, and audit data for activities e.g., distributed smallholder activities. It shall support geolocation, time-stamped photography, and unique identification of land parcels or trees.

Dynamic Reference Area (DRA)	An area of non-project land selected at the start of the project to serve as the simulated baseline. The DRA are statistically matched control sites that are chosen following the process defined in this methodology to create a "Synthetic Control" group. The boundaries are fixed ("Locked") during the Baseline Validity Period, but potential carbon stock changes are monitored dynamically via remote sensing during each crediting period.
Forest	As defined in the Forest Activity Requirements.
Leakage Belt	A defined geographical zone surrounding the Activity Area where activity shifting (displacement of pre-project land use) is monitored to quantify leakage emissions.
Module and Sub-modules	Modules are based on the key activity strategies as per the design of the activity based on which appropriate tracks are to be applied. There cannot be more than 1 Module within an activity design. Sub-modules can be added to a module as per the stacking eligibility requirements of the methodology.
Non-commercial plantation	Plantations that do not generate products for commercial trade and that do not meet the definition of a commercial plantation
Non-tree woody species	Woody species, for example, palms, bamboo and small shrubs and bushes, that typically do not have a distinct main trunk and often reach a height at maturity less than 2 meters.
Non-forest	An area of land that does not meet the definition of forest is treated as non-forest.
Organic Soil	Organic soils fulfil one of the following requirements: a. If the soil is never saturated with water for more than a few days, and contains >20% (by weight) of organic carbon (35% organic matter) b. If the soil is subject to water saturation episodes and has either: i. >12% (by weight) organic carbon (20% organic matter) if it has no clay ii. >18% (by weight) organic carbon (30% organic matter) if it has >60% clay iii. a proportional lower limit of organic carbon content between 12 and 18% if the clay content of the mineral fraction is between 0 and 60%
Sample Plot (PSP)	Sample plots are specific areas of a predefined shape and size within which carbon pools (biomass, dead organic matter, soil etc.) are measured. They are the basic units for data collection and shall be established and measured in a consistent manner to

	ensure that the data can be scaled up to represent the entire project area or stratum.
Project Area	As defined in Forest Activity Requirements.
Project Representative Farm	A representative area within smallholder farm selected randomly from the total population in Track 2 and Track 3 to be physically monitored for tree survival and growth, serving as a proxy for the broader cohort.
Scale of the activity	An activity (standalone project or a Voluntary Project Activity (VPA)) can be categorised based on the expected average removals per year and the project area. a. Micro scale: $\leq 10,000$ tCO₂e per year and not more than 500 ha b. Small scale: $\leq 60,000$ tCO₂e per year and not more than 10,000 ha) c. Large scale: $> 60,000$ tCO₂e per year or more than 10,000 ha)
Selective harvesting	Selective harvesting is a silvicultural system where individual trees or small groups of trees in mature stage are removed periodically from the forest stand for the main purpose of wood products, without clearing the entire area, in such a way that the rest of the trees are left standing. Selective harvesting helps in maintaining healthy, diverse forests over the long term. Selective harvesting is different from other forest management activities, such as thinning over-stocked forests for the purpose of increasing growth.
Smallholder project	As defined in the applicable Activity Requirements
Social Permanence	A risk mitigation concept applicable to Track 2, identifying that trees integrating livelihood benefits (e.g., fruit, fodder) have a lower risk of reversal due to the rational economic incentive for the farmer to retain them.
Spatially Explicit Matched Dynamic Baseline (SEMDB)	A baseline approach that compares the carbon stock change in the project area to the concurrent stock change in a statistically matched "Synthetic Control" group of pixels (The Dynamic Reference Area).
Synthetic Control	A statistical composite of non-project land parcels constructed to accurately reproduce the counterfactual trajectory of the treated unit (project area) in the absence of the intervention.
Tree	As defined in the Forest Activity Requirements
Tree Planting	As defined in the Forest Activity Requirements.

Wetland	As defined in the Forest Activity Requirements
Yield Maintenance	A leakage monitoring approach for smallholders where the yield of the primary subsistence crop is monitored to demonstrate that tree planting has not displaced food production.

3| SCOPE, APPLICABILITY, AND ENTRY INTO FORCE

3.1 | Scope

- 3.1.1.1 | This methodology applies to Afforestation, Reforestation, and Revegetation (ARR) activities that remove atmospheric CO₂ through the planting, sowing, or assisted natural regeneration of woody vegetation on land that is not defined as forest. The activity boundary covers the full lifecycle of the removals, including land preparation, planting, management, and the durable storage of carbon in living and dead biomass and soil pools.
- 3.1.2.1 | To accommodate diverse project management strategies and scales—from industrial timber plantations to smallholder agroforestry—this methodology employs a Modular Architecture. The specific rules for Baseline Setting (Section 7) and Monitoring (Section 15) are defined by the Compliance Track assigned to the activity based on the applicable Module.

3.2 | Applicability

The activity shall demonstrate compliance with the conditions below. Compliance is determined through a stepwise classification process.

- 3.2.1.1 | **Step 1: General Land & Species Eligibility:** The activity developer shall demonstrate that the land and species included in the activity boundary meet the following criteria regardless of the selected module:
- 3.2.1.1.1 | **Non-Forest Status:** The project area shall not meet the definition of a forest at the activity start date. Furthermore, the land shall have maintained non-forest status for at least 10 years prior to the activity start date.
- 3.2.1.1.2 | **No Conversion of Natural Ecosystems:** The land shall not have been converted from natural high-conservation-value ecosystems (e.g., wetlands, indigenous grasslands) to create the project area within the 10 years preceding the start date.
- 3.2.1.1.3 | **Species Characteristics:** The trees species shall be capable of reaching a minimum height of 2 meters at maturity. There is no minimum height at maturity for non-tree species.
- 3.2.1.1.4 | **Baseline - Natural Regeneration:** The activity developer shall demonstrate that environmental conditions (e.g., soil degradation, climate) or prevailing management activities (e.g., grazing pressure, cropping, wildfire) would

prevent the natural establishment of forest or non-tree species in the baseline scenario.

3.2.1.5 | **Wetlands Exclusion:** Planting areas shall not be located on wetlands.

3.2.1.6 | **Organic Soils (Peatlands):** Activities on organic soils are eligible only if:

- a. The area is not drained. Active drainage is prohibited, and existing drainage channels shall be blocked.
- b. The area is not irrigated, except for temporary irrigation required strictly for seedling establishment.
- c. Soil disturbance (e.g., ploughing, pit digging, stump removal) is limited to less than 10% of the specific planting area submitted for certification (not 10% of the entire project area).

3.2.1.7 | **Data Sufficiency (Track 1 Prerequisite):** Activities that are assigned to—or elect to use—**Track 1** shall demonstrate access to historical optical satellite imagery (e.g., Landsat 5/7/8, Sentinel-2) for the project region covering the full 10-year baseline historical period.

3.2.2. | **Step 2: Activity strategy:** The Activity Developer shall classify the project into one of the five modules and the sub-module based on the primary objective and scale, as explained in Section 1.2 | Strategy and Compliance Track Selection Matrix.

3.2.3. | **Step 3: Compliance Track Assignment:** Based on the module selected based on the strategy of the design, the activity developer shall demonstrate compliance to the specific Compliance Track (defined in Section 1.2 | Strategy and Compliance Track Selection Matrix).

3.2.4. | **Management Practices**

3.2.4.1 | The following management practices are permitted, subject to the restrictions of the assigned Track:

- a. **Rotation Forestry:** Harvesting is allowed with replanting².
- b. **Selective Harvesting:** Allowed across all strategy (and associated modules) provided it does not lead to conversion to non-forest land or forest degradation, or a reversal in the amount of carbon sequestered by the project over time. The project proponent is required to demonstrate that selective harvesting is sustainable, within the annual growth increment and will not lead to a sustained decrease in woody carbon stocks over time.
 - i. **Requirement:** A minimum tree density, as required by the applied forest definition, shall be retained across the project area at all times.

² Extension of rotation cycles, and accounting for avoided emissions are not allowed as per the methodology.

- ii. **Documentation:** The Project Design Document (PDD) shall explicitly detail harvesting prescriptions, including minimum age, target species, residual basal area, and minimum Diameter at Breast Height (DBH).

3.3 | Entry into Force

- 3.3.1. | The date of entry into force is 90 days from the publication date of this methodology.

3.4 | Mandatory Compliance and Safeguards

3.4.1. | Land Control and Ownership

- 3.4.1.1 | **Standalone Projects:** The activity developer shall demonstrate control over at least two-thirds (2/3) of the proposed project area at the time of Design Certification. Control over land refers to the land tenure requirements with reference to the applicable Activity Requirements, where legal ownership or rights to plant, manage and harvest trees can be demonstrated.

- 3.4.1.2 | **Carbon Rights:** The activity developer shall demonstrate clear legal ownership or rights to the carbon removals at the time of design certification and each performance certification. This includes formalizing land tenure for local rightsholders.

3.4.2. | Double Counting Mitigation

- 3.4.2.1 | **Double Issuance:** The activity developer shall follow the procedures in the [GHG Emissions Reduction & Sequestration Product Requirements](#).

3.4.2.2 | Program Overlap:

- a. No other ARR carbon programs shall exist within the same activity area i.e., land polygon.
- b. The activity developer shall ensure that the activity area is not registered under any other GHG program for the same carbon pools.

- 3.4.2.3 | **Regulatory Conflict:** The activity shall not conflict with any national, sub-national, or local regulations or guidance.

3.4.3. | Environmental Safeguards:

- 3.4.3.1 | **Invasive Species:** Species selected for planting shall not be classified as invasive in the Host Country.

- 3.4.3.2 | **Biodiversity:** For Commercial forestry of all scales (Module 1), the activity shall dedicate at least 10% of the project area to native vegetation conservation or restoration corridors to ensure ecological connectivity following the HCV approach.

- 3.4.3.3 | **Resource Diversion:** Precautionary Principle - Activities shall adhere to Safeguarding Principle 9 (Environment, Ecology, and Land Use), ensuring a precautionary approach to avoid negative environmental impacts.

3.4.4. | **Social Safeguards (Food Security & Displacement):**

3.4.4.1 | **Displacement:** Activities shall not lead to the displacement of the production of goods (e.g., food, feed) under the baseline scenario.

3.4.4.2 | **Leakage:** If displacement occurs, the project developer shall apply the Gold Standard leakage requirements. For Modules 3 and 4 (Production and Subsistence Agroforestry respectively), if trees displace more than 10% of the arable crop area, the activity shall demonstrate that the total farm productivity (Land Equivalent Ratio) is maintained or enhanced.

3.4.5. | **Durability and Permanence**

3.4.5.1 | **Baseline Validity Period:** The boundaries of the Reference Area (for Tracks 1 and 2) are fixed ("Locked") for the duration of the Baseline Validity Period (**10 years**).

3.4.5.2 | **Reversal Management:** In accordance with the [GHG Emissions Reduction & Sequestration Product Requirements](#), 20% of issued GS-VERs shall be transferred to the Gold Standard buffer to manage reversal risks.

3.4.5.3 | Any unforeseen reversal (*force majeure*) after the end of the crediting period shall be compensated through retiring credits from the Gold Standard common buffer pool, as long as the post crediting monitoring data is presented to Gold Standard.

4 | **SOURCES AND REFERENCES**

4.1.1. | This methodology refers to following methodologies, tools, and documents:

4.1.1.1 | Gold Standard:

- a. [Forestry Activity Requirements](#).
- b. [Requirements for Additionality Demonstration](#).
- c. [Requirements for baseline determination in methodologies](#).
- d. [Requirements for addressing leakage in methodologies](#).
- e. [Performance Shortfall Guidelines, Requirements and Procedure](#).
- f. [GHG Emissions Reduction & Sequestration Product Requirements](#).
- g. [Risks and Capacities Guidelines for Agriculture and Forestry](#).
- h. [Tool 5 – Downward Adjustment Factor Determination](#).

5 | **ACTIVITY BOUNDARY AND GHGS SOURCES/SINKS**

5.1 | **Activity Boundary**

5.1.1. | The activity boundary encompasses all anthropogenic sources of greenhouse gases (GHGs) and carbon sinks that are under the control of the activity developer, are related to the activity, or are significantly affected by it. The

boundary follows a lifecycle approach, comprising the physical sites and transport links associated with land preparation, planting, and durable carbon storage.

5.1.2. | **Spatial Boundary:** The activity developer shall clearly identify the spatial extent of the activity using the following three distinct designations:

- a. **Activity Area:** The physical geographical locations where vegetation is established and monitored and verified for SDG impacts.
 - i. **Delineation:** Each discrete parcel in the Activity Area shall be uniquely identified using Global Positioning System (GPS) coordinates and vector maps (shapefiles/polygons).
 - a. **Smallholder Provision:** The boundary may be defined as the perimeter of the participating farm, provided each tree is recorded using the Census method and associated GPS coordinates.
 - ii. **Exclusion:** The boundary shall explicitly exclude wetlands, existing forest patches, indigenous grasslands, ineligible riparian zones, roads and other infrastructure.
 - iii. **Control:** The spatial boundary exists only within areas under the control of the activity participants. For standalone activities, the activity developer shall demonstrate control over at least two-thirds (2/3) of the proposed activity area at the time of Design Certification. Control over 100% of the area and the final project boundary shall be confirmed at the first performance certification.
- b. **Reference Area Boundary:** To support the Dynamic Baseline requirements of **Track 1** and **Track 2**, the activity developer shall define a spatial domain from which baseline data is derived.
 - i. **Applicability:** Mandatory for Track 1 and Track 2.
 - ii. **Definition:** A set of control parcels or administrative zones located outside the planting area but within the same climatic and geomorphic region.
 - iii. Once validated, the boundary of the Reference Area is fixed ("locked") for the duration of the Baseline Validity Period (10 years). While the carbon stock data within these boundaries is monitored at least once every 5 years or before each verification to track climate variability, the location of the reference area does not change.
- c. **Leakage Buffer Zone** The geographical area surrounding the Planting Area where activity shifting (displacement) is most likely to occur.
 - i. **Requirement:** A spatial buffer surrounding each discrete planting polygon (e.g., **2 km** for activities displacing arable agriculture; **5 km** for grazing).

5.1.3. | **Temporal Boundary:** The temporal boundaries are defined by the Crediting Period (40 to 100 years) and the Baseline Validity Period (10 years).

5.2 | Greenhouse Gas Sources and Carbon Pools

5.2.1. | **Carbon Sinks (Removals):** The selection of carbon pools depends on the Activity Module and Compliance Track. The activity developer shall account for pools as indicated in Table 7.

Table 6: Carbon Sinks

Carbon Pool	Description	Included	Justification
Aboveground Biomass	Living biomass in stems, branches, bark, and leaves.	Yes	Primary source of carbon removal.
Belowground Biomass	Living biomass in roots.	Yes	Significant pool for carbon storage. Calculated using root-to-shoot ratios.
Dead Wood	Standing or lying dead wood.	Optional	Recommended for Module 2 (Restorative forestry). Optional for others if <i>de minimis</i> . Optical remote sensing data cannot be used.
Litter	Leaves and twigs on soil surface.	Optional	Excluded for Module 3 (Production Agroforestry) and Module 4 (Subsistence Agroforestry) for conservativeness. Optional for others.
Soil Organic Carbon (SOC)	Organic carbon in mineral soil.	Optional	Optional for all Modules. In Module 3 or 4 SOC is restricted only if the agriculture activity leads to conservation of SOC.
Non-woody Biomass	Grass, herbs, and other non-woody plants.	No	Excluded to maintain conservativeness and simplify monitoring.
Long lived Wood Products (LLWP)	Timber/Pulp stored in long-term products.	Optional	Applicable only to Module 1 and Module 3.

5.2.2. | **Baseline emissions/removals:** The baseline scenario represents the carbon flux that would have occurred in the absence of the activity (e.g., continued

degraded grassland or marginal cropping). To ensure conservativeness, most baseline emissions are excluded (set to zero), meaning the activity does not claim credit for "avoided emissions" (e.g., avoided burning), only for baseline removals.

Table 7: GHGs included in the Baseline Scenario

Source	Description	GHGs	Included?	Justification
Baseline Removals <i>Existing Vegetation</i>	Pre-project biomass and SOC.	CO ₂	Yes	The Core Baseline. Represents the ongoing sequestration (or loss) of pre-project woody vegetation (e.g., woody vegetation/shrubs) captured via monitoring the reference area.
Biomass Burning	Natural or human-caused fires.	CO ₂	No	Excluded. Carbon stock changes due to fire are captured in the remote sensing of the reference area.
		CH ₄ , N ₂ O		Excluded for conservativeness (omitting baseline emissions lowers the net benefit).
Fertilizer Use <i>(Baseline Agriculture)</i>	Nitrogen fertilizer application.	CO ₂ , CH ₄ , N ₂ O	No	Excluded for conservativeness.
Fossil Fuel Use <i>(Baseline Machinery)</i>	Burning of fuels for baseline operations.	CO ₂ , CH ₄ , N ₂ O	No	Excluded for conservativeness.

5.2.3. | **Activity emissions/removals:** Activity emissions are those resulting directly from the implementation of the ARR activity. The activity developer shall account for these as "Activity Emissions" to be deducted from the gross removals.

Table 8: Sources of Activity Emissions

Source	Description	GHGs	Included?	Justification
Biomass Loss	Biomass loss owing to	CO ₂	Yes	The primary metric.

(Harvesting / Management)	harvesting or other forest management practices.			Biomass loss shall be monitored primarily through full-area census measurements. Where a complete census is not feasible due to operational, logistical, or ecological constraints, biomass loss may be estimated using statistically robust sample plots that are representative of the monitored area. Accounted for as a decrease in carbon stocks.
Biomass Burning	Fires during site preparation or management.	CO ₂	No	Accounted for as a carbon stock change.
		CH ₄ , N ₂ O	Yes	Mandatory. Significant source of non-CO ₂ emissions.
Fertilizer Use	Application of nitrogen fertilizers. (Planting / Nursery)	CO ₂ , CH ₄ ,	No	In-significant source of emissions.
		N ₂ O	Conditional	Included if synthetic nitrogen fertilizers are applied. Excluded if only organic compost/manure is used or if application is zero.
Fossil Fuel Use	Machinery for site preparation or transport.	CO ₂	Conditional	Track 1: Mandatory if emissions >5% of removals. Track 2 & 3: Optional/Excluded (assumed <i>de minimis</i>).
		CH ₄ , N ₂ O	No	Excluded. In-significant source of emissions.
Livestock Emissions	Enteric fermentation and manure management.	CH ₄	Conditional	Module 3 and 4). Included only if the number of head of livestock increases

(Silvo-
pasture
Only)

within the activity
boundary compared to
the baseline.

Question for Stakeholders:

Livestock emissions

Livestock methane emissions from integrated land-use modules are proposed to be accounted. The methodology requires for accounting of methane emissions from increase in number of head of livestock in the project area. This is to ensure that intensified livestock management does not undermine the project's net carbon sequestration.

Feedback Requested: We invite input on appropriate monitoring thresholds and whether IPCC Tier 1 or Tier 2 modeling should be required for incidental grazing within these diverse scenarios.

5.2.4. | **Sources of Leakage Emissions:** Leakage is the increase in GHG emissions outside the Activity Boundary attributable to the activity.

Table 9: Sources of Leakage Emissions

Source	Description	GHGs	Included?	Justification
Activity Shifting	Displacement of pre-project agricultural or grazing activities.	CO ₂	Yes	Mandatory. Displacement of pre-project agricultural or grazing activities into the Leakage Buffer Zone or broader forest area.
		CH ₄ , N ₂ O	Conditional	Included only if the displaced activity involves fire (e.g., slash-and-burn agriculture).
Market Effects	Changes in supply or demand of forestry products.	CO ₂	Conditional	Applicable only to Commercial Forestry (Module 1) or Production Agroforestry (Module 3) if the reduction in timber supply causes increased harvesting elsewhere. Excluded for Module 2, 4, and 4.

6| DEMONSTRATION OF ADDITIONALITY

6.1 | Requirements

- 6.1.1. | The Activity Developer shall demonstrate that the ARR activity would not have occurred in the absence of the incentives provided by carbon finance. The demonstration of additionality shall be conducted in accordance with the latest version of the [GS4GG Methodology Standard: Requirements for Additionality Demonstration](#).
- 6.1.2. | The specific requirements for demonstrating additionality are determined by the Compliance Track assigned in Section 3.
- 6.1.3. | **Mandatory Analyses:** All activities, regardless of the Compliance Track selected, shall successfully complete the following mandatory analyses:
- Regulatory Surplus Analysis (Section 6.3)
 - Lock-In Risk Analysis (Section 6.4)
 - Ongoing Financial Need (at Renewal) (Section 6.8)
- 6.1.4. | **Additionality Pathway Selection:** Following the mandatory analyses, the Activity Developer shall demonstrate additionality through one of the options defined in Table 11, determined by the Compliance Track and Module.

Table 10: Additionality Pathway Selection Matrix

Module	Permitted Demonstration Approaches
Commercial Forestry (Module 1), Large-scale Production Agroforestry (Module 3, Track 1), and Large-scale Revegetation (Module 5, Track 1)	Option C: Financial Analysis (Investment analysis Mandatory) <i>Rationale: These scale and design types generate significant timber or product revenue. Barrier Analysis is prohibited as a standalone demonstration to prevent non-additional commercial projects</i>
Restorative Forestry (Module 2), Small/Micro-scale Production Agroforestry (Module 3, Tracks 2 & 3), Subsistence Agroforestry (Module 4), and Small/Micro-scale Revegetation (Module 5, Tracks 2 & 3) with no stacking	Option A: Deemed Additionality (Positive List) – <i>If eligible</i> Option B: Barrier Analysis Option C: Financial Analysis (Investment analysis)
Revegetation (where Module 5 is stacked with Modules 1, 2, 3 or 4) and Smallholder Aggregations (Sub Module A)	Shall follow the permitted approaches for core module where Module 5 or submodule A are added.

6.2 | Option A: Deemed Additionality (Positive List)

6.2.1. | In accordance with Section 5.4 of the [GS4GG Common Practice Tool](#), activities meeting the category specific criteria below are **Deemed Additional** and are exempt from conducting a separate Financial or Barrier Analysis, provided they pass the Mandatory Screening (6.3 & 6.4).

6.2.2. | Category 1: LDCs and SIDS (Location-Based)

- a. **Criteria:** The activity is located in a Least Developed Country (LDC) having an HDI less than 0.55, Small Island Developing State (SIDS), or Landlocked Developing Country (LLDC).
- b. **Scale:** The activity meets the definition of Micro-Scale or Small-Scale.
- c. **Constraint:** The activity is **not** Commercial Forestry or Production Agroforestry (Module 1 or Module 3).
- d. **Justification:** Activities in these regions face disproportionate investment risks and institutional barriers recognized by GS4GG.

6.2.3. | Category 2: Non-Commercial Restoration (Activity-Based)

- a. **Criteria:** The activity involves the restoration of Native or High Conservation Value (HCV) ecosystems where:
 - i. Restorative Forestry (Module 2) is applied; AND
 - ii. The activity developer provides a binding legal instrument (e.g., conservation easement or statutory designation) prohibiting commercial timber harvesting for the duration of the crediting period.
 - iii. It shall be demonstrated objectively with supporting evidence and maps that the core activity involves restoration of Native or High Conservation Value (HCV) ecosystems.
- b. **Justification:** The conservation only activities generate no internal rate of return (IRR) and face inherent financial barriers without carbon finance.

6.3 | Regulatory surplus analysis (Mandatory)

6.3.1. | The activity developer shall demonstrate that the activity is not mandated by any existing laws, regulations, statutes, or binding mandates (legal requirements) within the Host Country's jurisdiction. The analysis shall verify the following:

- a. **Host Country Eligibility:** The project activity type shall not be excluded or declared ineligible by the host country (e.g., via a negative list of activities, technologies, or measures for the issuance of carbon credits). If no such list is available from the host country, the activity shall be assumed as not excluded or declared ineligible.
- b. **Legal Mandates:** The project activity shall not be mandated by any existing or pending law, statute, regulation, standard, or legal

requirement within the host Party's jurisdiction. An exception is permitted if the law or regulation explicitly refers to or formally integrates Article 6 based mechanisms as an instrument for implementation. Evidence shall be provided demonstrating this exemption.

- i. **Mandate Reforestation:** The law shall not explicitly require the planting of trees on the specific land parcels (e.g., as a condition for a mining concession permit, compensation for deforestation elsewhere, or post-harvest replanting obligations).
- ii. **Prohibit Baseline Land Use:** The law shall not explicitly prohibit the baseline scenario (e.g., grazing on degraded land) and systematically enforce this prohibition.
- c. **Systematic Non-Enforcement:** If a legal mandate exists (e.g., a "National Forest Law" requiring tree cover maintenance) but is not systematically enforced, the activity is considered to have Regulatory Surplus.
 - i. **Threshold:** The activity developer shall provide evidence (e.g., government reports, remote sensing data, or peer-reviewed literature) demonstrating that non-compliance is "Common Practice" (> 50% non-compliance rate) in the relevant region.

6.4 | Lock-In Risk Analysis (Mandatory)

- 6.4.1. | In accordance with Section 6.2 of the Additionality Standard, the activity developer shall demonstrate that the activity does not "lock in" high-emitting technologies or practices incompatible with the long-term goals of the Paris Agreement.
- 6.4.1.1 | **Technology Lock-In:** ARR activities generally mitigate lock-in risks by sequestering carbon. However, for Module 1 (Commercial Forestry), the activity developer shall demonstrate that the activity does not lock in short-rotation monocultures that degrade soil health or biodiversity, thereby increasing future vulnerability to climate change (e.g., pests, fire).
- 6.4.1.2 | **Economic Lock-In:** The activity shall not create a dependency on fossil-fuel-intensive supply chains (e.g., excessive synthetic fertilizer transport) that would persist beyond the crediting period.

6.5 | Option B: Barrier Analysis

- 6.5.1. | **Applicability:** Permitted for modules and tracks specified in Table 11 that do not qualify for the Positive List.
- 6.5.2. | The activity developer shall demonstrate that at least one significant barrier prevents the implementation of the activity and that carbon finance is the decisive factor in overcoming it.

6.5.2.1 | **Eligible Barriers:** The eligible barriers considered within the scope of this methodology are:

- a. **Investment Barrier:** Inability to access debt financing due to lack of collateral (common for smallholders) or prohibitive interest rates (e.g.: > 15%) in the Host Country.
- b. **Institutional Barrier:** Absence of secure land tenure or weak enforcement of property rights requires the project to fund extensive land titling, governance structures, or aggregation of fragmented parcels.
- c. **Technological/Ecological Barrier:** The project area is severely degraded (e.g., mining spoil, saline soils) requiring specialized, high-cost inputs (e.g., soil amendments, specific pioneer species) that are not financially viable under standard land-use models.

6.5.2.2 | **Evidence:** Barriers shall be substantiated by independent, verifiable evidence (e.g., bank rejection letters, official land tenure assessments, soil analysis reports) and not anecdotal claims.

6.6 | Option C: Financial Viability (Investment Analysis)

6.6.1. | **Applicability:** Mandatory for Commercial Forestry (Module 1) and large-scale Production Agroforestry (Module 3, Track 1) and large-scale Revegetation (Module 5, Track 1). Optional for others.

6.6.2. | The activity developer shall conduct an Investment Analysis following Section 6.3 of the Additionality Standard.

6.6.2.1 | Approach:

- a. **Investment Comparison Analysis:** Used if the baseline is a Production commercial production activity. The activity developer shall demonstrate that the ARR activity is less financially attractive than the baseline land use without carbon revenues.
- b. **Benchmark Analysis:** Used if the ARR activity is a standalone commercial venture. The activity developer shall demonstrate that the Project IRR (without carbon) is lower than the relevant Sectoral Benchmark (Hurdle Rate) or Weighted Average Cost of Capital (WACC).
- c. **Critical Parameters:** The financial model shall account for all operational revenues and costs, including:
 - i. **Revenue:** Projected timber/wood product sales, fruit/crop revenues, thinnings, and subsidies (excluding carbon).
 - ii. **Costs:** Land acquisition/lease, site preparation, seedlings, labour, management, and taxes.
- d. **Sensitivity Analysis:** The conclusion shall hold robust even if key parameters (e.g., timber price, yield, emission reduction Volume, GSVER price etc.) deviate by $\pm 10\%$.

6.7 | Common practice analysis

- 6.7.1. | The Activity Developer shall demonstrate that the proposed activity is not "Common Practice" in the relevant geographical area.
- 6.7.2. | **Methodology-Level Assessment (Tracks 2 & 3):** Valid until 31 December 2028. Based on FAO Global Forest Resources Assessment data, non-commercial restoration and revegetation activities implemented without carbon finance represent a negligible fraction (<5%) of land-use change globally. Therefore, activities under Restorative Forestry (Module 2) and Subsistence Agroforestry (Module 4) and Revegetation (Module 5) and Smallholder aggregation (Sub-module A) are Deemed Not Common Practice, provided they are not mandated by law.
- 6.7.3. | **Activity-Level Assessment:** For Commercial Forestry and Production Agroforestry or earmarked commercial Revegetation, the activity developer shall conduct a specific Common Practice Analysis:
- 6.7.3.1 | **Geographic Scope:** Define the applicable region (e.g., eco-region or sub-national jurisdiction).
- 6.7.3.2 | **Similar Activities:** Identify existing commercial forestry activities of similar scale, species, and regulatory environment operating without carbon finance.
- 6.7.3.3 | **Essential Distinction:** If "Similar Activities" represent >20% of the sector, the project is deemed Common Practice and is non-additional unless an Essential Distinction is proven (e.g., the project targets highly degraded land where commercial yields are significantly lower than standard plantations).

6.8 | Ongoing Financial Need (OFN)

- 6.8.1. | At the renewal of the crediting period, the additionality of the activity shall be reassessed. The activity developer shall demonstrate Ongoing Financial Need in accordance with Section 7 of the Additionality Standard.
- 6.8.1.1 | **Requirement:** The developer shall provide an updated financial analysis or qualitative assessment demonstrating that carbon revenues are still required to cover operational costs (e.g., monitoring, guarding) or to overcome persistent market barriers.
- 6.8.1.2 | **Graduation:** For Commercial Forestry, if the activity has achieved commercial viability ($IRR > \text{Benchmark}$) through timber revenues alone due to market maturation, it is no longer additional. The activity shall be ineligible for crediting period renewal for new growth, though it shall maintain existing carbon stocks for the duration of the permanence monitoring period.

7| BASELINE SCENARIO AND EMISSIONS

7.1 | Selection of baseline approaches

7.1.1.1 | The Crediting Baseline ($BL_{crediting,y}$) shall be determined following the mandatory stepwise approach defined in the GS4GG Methodology Standard: Requirements for Baseline Determination:

- a. Step 1: Selection and Justification of the Baseline Approach.
- b. Step 2: Determination of the Unadjusted Baseline (BL_{unadj}).
- c. Step 3: Application of the Downward Adjustment (DAF).
- d. Step 4: Identification of a Conservative Business-As-Usual (BAU) Baseline.
- e. Step 5: Comparison and Selection of the Final Crediting Baseline.

7.2 | Step 1: Selection and Justification

7.2.1.1 | **Selection of Baseline Approach:** This methodology utilizes approach (c) from paragraph 5.6.2 of the Methodology Standard - Requirements for Methodology Development: An approach based on existing actual or historical emissions, adjusted downwards/ upwards.

7.2.2.1 | **Justification for the Baseline Approach:**

7.2.2.1.1 | **Correction for Environmental Forcing:** Natural regeneration in non-forest lands is stochastic, driven heavily by precipitation variability and climate change (CO_2 fertilization). Static baselines cannot distinguish between management-induced growth and climate-induced growth.

7.2.2.2.1 | **Difference-in-Differences (DiD):** By monitoring actual conditions in a Reference Area rather than utilizing hypothetical projections, the methodology isolates the project effect. This ensures credits are issued only for the Net Difference in carbon stock change between the project and the baseline, removing counterfactual bias.

7.3 | Step 2: Determination of Unadjusted Baseline

7.3.1.1 | The Unadjusted Baseline ($BL_{unadj,y}$) represents the net change in carbon stocks observed in the absence of the activity.

7.3.2.1 | **General Architecture:** For Track 1 and Track 2, the baseline employs a strict separation between the "Spatial Domain" and the "Temporal Data" to ensure regulatory certainty.

7.3.2.1.1 | **Year 0 (The Lock):** At the time of Design Certification, the geographic boundaries of the Reference Area (RA)—the control cohort—are identified based on the matching criteria. These spatial boundaries are fixed ("Locked") for the duration of the Baseline Validity Period (10 years).

7.3.2.2 | Years 1–10 (The Monitor): While the reference area remains fixed, the data extracted from it is dynamic. The baseline is calculated based on remote sensing observations of the reference area in year y .

7.3.2.3 | Renewal Audit: At the end of the 10-year validity period, the Reference Area is audited. Land that has undergone significant land-use change unrelated to natural regeneration (e.g., urbanization, conversion to cropland) shall be removed and replaced following the procedure in Annex 1, and the map is re-locked for the subsequent renewal period.

7.3.3. | Track 1 Method: Spatially Explicit Matched Dynamic Baseline (SEMDB)

- a. **Applicability:** Mandatory for Commercial forestry and large-Scale Production agroforestry, optional for other modules.
- b. **Concept:** The baseline is derived from a "Synthetic Control" group - a specific cohort of non-project pixels that are biophysically and socio-economically identical to the project area.
- c. **Procedure:** The activity developer shall construct the Synthetic Control group following the detailed statistical procedures in [Annex 1](#). The selection shall adhere to the Mandatory Covariate Thresholds defined in Table 12 below.

7.3.4. | Procedure:

7.3.4.1 | Stage A: Candidate Pool Generation

- a. Identify the "Donor Pool" of all non-forest land within the host region (e.g., Level-3 Ecoregion) that share the project's broad land-use class.
- b. Land within the Leakage Buffer Zone (minimum 5km radius) are excluded to prevent contamination.

7.3.4.2 | Stage B: Covariate Matching

- a. Estimate the parameters from Table 11 for the Reference and Project Areas. Select specific control pixels from the Donor Pool using an appropriate Propensity Score Matching algorithm. Matches are valid only if they fall within the tolerances specified in Table 11. Additional parameters are permitted with appropriate justification.

7.3.4.3 | Stage C: Statistical Validation

- a. **Covariate Balance Check:** The activity developer shall demonstrate that the Standardized Mean Difference (SMD) between the mandatory parameters measured in the Project Area and the chosen Reference Area for all mandatory covariates is < 0.1 . Matches failing this test are invalid.
- b. **Parallel Trends Assumption:** The activity developer shall demonstrate that the trend of biomass change in the Project and Reference areas was parallel for at least 5 years prior to the project start.

7.3.4.4 | Stage D: Calculation

- a. The unadjusted baseline is calculated as the mean Carbon Stock Change ($\Delta C_{ref,y}$) of the Locked Reference Area for year y .

$$BL_{unadj,y} = \Delta C_{ref,y} \times A_{project} \quad (\text{Eq-1})$$

Where

Parameter	Description	Units
$BL_{unadj,y}$	Unadjusted Baseline Removals in year y	tC
$\Delta C_{ref,y}$	Mean Carbon Stock change of the Locked Reference Area	tC/ha/yr
A_i	Total Project Area	ha

Table 11: Mandatory Covariates for Baseline Matching

Category	Covariate (Metric)	Matching Tolerance / Criteria
Edaphic (Soil)	Soil Texture	Shall match the Texture Class (e.g., Sandy Loam vs. Clay).
Topography	Elevation	Matches shall be within ± 200 meters of project elevation.
	Slope	Matches shall be within ± 10 degrees of project slope.
Climatic	Precipitation	Shall be located within the same Level-3 Ecoregion (Isohyet).
Veg. History	Pre-Project Vegetation	There shall be no significant statistical difference ($p > 0.05$) between the Mean NDVI ³ /EVI for the 5 years prior to the start date between the reference and project areas.
Tenure	Legal Status	Shall share the same Tenure Status (e.g., Communal vs. Private).

7.3.5. | Track 2 Method: Regional Dynamic Aggregate

- Applicability:** Based on the module and scale considered in the project design.
- Concept:** To ensure inclusivity for small and microscale projects that lack the capacity for GIS matching, this track uses a "Subscription Model."
- Procedure:**

³ NDVI = Normalised Differential Vegetation Index.

- i. **Selection:** The activity developer shall "subscribe" to the Regional Dynamic Curve approved by Gold Standard or a recognized national authority for the specific Eco-Zone and Land Class (e.g., "Degraded Miombo Savanna").
 - ii. **Aggregated NFI Fallback:** If a published curve is unavailable, the developer may derive the rate from National Forest Inventory (NFI) plots. **Requirement:** Data shall be aggregated from at least 30 plots within the same administrative district and updated within the last 5 years.
- d. **Calculation:**

$$BL_{unadj,y} = \Delta C_{regional,y} \times A_{project} \quad (\text{Eq-2})$$

Where

Parameter	Description	Units
$BL_{unadj,y}$	Unadjusted Baseline Removals in year y	tC
$\Delta C_{regional,y}$	Mean Carbon Stock change of the Reference Region	tC/ha/yr
A_i	Total Project Area	ha

7.3.6. | Track 3 Method: Static Historical Trend

- a. **Applicability:** Based on the module and scale considered in the project design.
- b. **Procedure:** Due to the difficulty of distinguishing non-tree biomass growth from seasonal noise via satellite, a conservative Static Baseline is applied.
 - i. **Trend Analysis:** Analyse the vegetation index (e.g., NDVI) for the 10 years prior to the start date.
 - ii. **Forward Projection:** Project the linear trend forward.
 - i. **Default:** If the historical trend is negative (degrading), the baseline is assumed to be zero.
 - ii. **Cap:** The projected baseline biomass shall not exceed the carrying capacity of the regional shrubland class (defined in Annex 1).

7.3.7. | Safeguard: Hybrid Monitoring

- a. **Applicability:** Tracks 1 & 2.
- b. **Issue:** Optical satellite sensors cannot reliably detect biomass changes in seedlings or saplings (<1.5m). Comparing "Project Bare Soil" to "Reference Bare Soil" in early years creates a blind spot where the dynamic baseline may erroneously report zero growth.

- c. **Mitigation:** The determination of $BL_{unadj,y}$ shall follow a Hybrid Phase approach as detailed in Annex 1:

7.3.7.1 | **Phase 1 (Establishment, Years 0-3):** The dynamic comparison is suspended. The baseline removal is deemed zero for the purpose of dynamic comparison.

- a. **Exception 1:** If the Reference Area shows a spectral spike (>20% increase in NDVI) indicating rapid disturbance recovery (e.g., post-fire regrowth), the "Zero" assumption is void, and the observed reference rate shall be deducted.
- b. **Exception 2:** In high latitude/elevation or arid/semi-arid areas, it may take more time to reach the desired height. In such cases, the expected year of canopy closure shall be defined at the design certification stage and shall be endorsed by the VVB. Year of canopy closure shall not go beyond year 9 in such cases.

7.3.7.2 | **Phase 2 (Minimum canopy Closure, Year 4+):** Once project trees reach detectable height (>1.5m) or canopy closure (>10%), the full Dynamic Comparison activates.

7.4 | Step 3: Application of Downward Adjustment (DAF)

7.4.1.| To ensure ambition and account for measurement errors, a Downward Adjustment Factor (DAF) is applied in accordance with [Tool 5 – Downward Adjustment Factor Determination](#) (GS4GG PAA MT400-05).

7.4.2.| **Addressing Uncertainty:** Uncertainty is addressed through parameter-level conservativeness in the quantification steps and is not aggregated into the DAF.

- a. **For Track 1:** Calculated based on the Standard Error (SE) of the Synthetic Control matches at the 90% Confidence Interval (See Annex 1).
- b. **Track 2 & 3:** A default uncertainty deduction of 10% is added to the DAF to account for the lack of bespoke matching. This represents the UNC_{BSL} parameter.

7.4.3.| **Ambition Component:**

- a. **Default Floor:** The Absolute DAF Floor of 1.25% applies for Land Use and Forestry (LUF) activities.
- b. **Sectoral Override:** If the Host Country has a specific Sectoral DAF defined in the [Tool 5 – Downward Adjustment Factor Determination](#) that is higher than 1.25%, the higher value shall apply.
- c. **Sectoral Override:** [Tool 5 – Downward Adjustment Factor Determination](#) **Application Formula:** For Removal activities, the fixed DAF and uncertainty are applied to increase the baseline deduction (calculated in Section 7.9) following Equation 4 of the DAF Tool.

7.5 | Step 4: Identification of Conservative BAU Scenario

- 7.5.1. | The activity developer shall identify a "Crediting Cap" based on plausible alternative land uses.
- 7.5.2. | **Regulatory Targets:** Identify if there are any national or sub-national reforestation targets (e.g., NDC unconditional targets) applicable to the region. Convert this target into a per-hectare rate ($Rate_{Gov}$).
- 7.5.3. | **"Avoided Degradation" Safeguard:** This methodology credits Net Removals, not Avoided Emissions. If the Reference Area (Dynamic Baseline) exhibits a decrease in carbon stocks (Degradation) during the monitoring period (i.e., $\Delta C_{ref,y} < 0$), the Unadjusted Baseline shall be set to Zero.

7.6 | Step 5: Selection of Crediting Baseline

- 7.6.1. | The Final Crediting Baseline ($BL_{crediting,y}$) is mathematically determined in Section 7.9 to ensure the final deduction reflects the maximum conservative value (accounting for DAF and Government Targets).

7.7 | Calculation of Baseline Removals

- 7.7.1. | The activity developer shall calculate the Baseline Removals (BR_y) for the monitoring period y . Baseline Removals represent the quantity of carbon dioxide (tCO₂e) that would have been sequestered by natural regeneration within the Activity Boundary in the absence of the project activity. The activity is credited only for the difference between the Activity Removals (AR_y) and the Baseline Removals BR_y calculated in this section.
- 7.7.2. | **Baseline Removals (BR_y):** Baseline removals are the stock changes in living biomass (tree and non-trees woody biomass) and dead organic matter (dead wood and litter) that would have occurred in the absence of the project activity during the monitoring period y . The activity developer shall calculate the Unadjusted Baseline Removals ($BR_{unadj,y}$) using the specific Baseline Removal Rate (ΔC_{BSL}) determined in Section 7 (Step 2), applied to the eligible area.

Step 1: Determine the Applicable Rate

- 7.7.2.1 | The source of the rate depends on the assigned Compliance Track (Section 3) and Monitoring Phase ([Annex 1](#)).

Table 12: Baseline Removal Rate Sources

Compliance Track	Monitoring Phase	Source of Rate (ΔC_{BSL})	Value Logic
Track 1 (Dynamic SEMDB)	Phase 1: Establishment (Years 0–3)	Annex 1 (A1.5)	0 (Default). <i>Exception: If Reference Area monitoring detects a disturbance spike (VI increase >20%), the observed rate applies.</i>

	Phase 2: Canopy Closure (Year 4+)	Section 7.3.2	$\Delta C_{ref,y}$ The mean carbon stock change (tC/ha) observed in the "Locked" Reference Area pixels for year y .
Track 2 (Regional Dynamic)	Phase 1: Establishment (Years 0–3)	Annex 1 (A1.5)	0 (Default).
	Phase 2: Minimum Canopy Closure (Year 4+)	Section 7.3.3	$\Delta C_{regional,y}$ The rate derived from the approved Regional Dynamic Curve or Aggregated NFI (tC/ha).
Track 3 (Static Exception)	All Years	Section 7.3.4	Linear Trend The rate projected from the historical 10-year vegetation trend (tC/ha).

Step 2: Apply the "No Avoided Degradation" Safeguard

7.7.2.2 | If the determined rate ΔC_{BSL} is negative (indicating the Reference Area lost biomass/degraded during the period), the rate shall be set to Zero for the calculation of $BR_{unadj,y}$. This methodology credits Net Removals. It does not issue credits for "Avoided Degradation" (avoided baseline emissions).

Step 3: Calculate Unadjusted Removals

7.7.3. | The removals are calculated as:

$$BR_{unadj,y} = \sum_{i=1}^M (\Delta C_{BSL} \times A_i \times T_y) \times \frac{44}{12} \quad (\text{Eq-3})$$

Where

Parameter	Description	Units
$BR_{unadj,y}$	Unadjusted Baseline Removals in year y (prior to DAF application)	tCO ₂ e
i	Stratum index (e.g., land class).	Dimensionless
M	Total number of strata	Dimensionless
ΔC_{BSL}	Baseline Removal Rate for stratum i	tC/ha/yr
A_i	Total eligible area of stratum i	ha
T_y	Duration of the monitoring period	years
$\frac{44}{12}$	Conversion factor from Carbon (C) to Carbon Dioxide (CO ₂).	Ratio

7.8 | Baseline Emissions ($BE_{MU,t}$)

- 7.8.1. | **Quantification of Baseline Emissions:** Baseline Emissions (BE_y) refer to the GHG emissions (e.g., N_2O from fertilizer, CH_4 from grazing or biomass burning) that would have occurred in the absence of the activity. To ensure conservativeness, baseline emissions are excluded from the crediting equation.

$$BE_y = 0 \quad (\text{Eq-4})$$

Where

Parameter	Description	Units
BE_y	Baseline Emissions in year y	tCO ₂ e

- 7.8.2. | **Difference between BAU and baseline emissions or removals:** This section identifies the Business-As-Usual (BAU) Floor to ensure the project does not credit removals that are legally mandatory.

- 7.8.2.1 | The activity developer shall identify if a Host Country government target (e.g., unconditional NDC reforestation target or legal replanting mandate) exists for the region.

- a. **Calculation:** Convert the government target into a removal quantity ($BR_{gov,y}$).

$$BR_{gov,y} = \sum_{i=1}^M \left(Rate_{gov} \times A_i \times T_y \times \frac{44}{12} \right) \quad (\text{Eq-5})$$

Where

Parameter	Description	Units
$BR_{gov,y}$	Baseline removals mandated by government targets	tCO ₂ e
$Rate_{gov}$	Mandated reforestation or carbon accumulation rate for stratum i	tC/ha/yr
A_i	Total eligible area of stratum i	ha
T_y	Duration of the monitoring period	years

- b. **Comparison:** This value serves as the minimum deduction. The final Crediting Baseline cannot be lower than $BR_{gov,y}$.

7.9 | Application of downward adjustment

- 7.9.1. | To align with the Paris Agreement and Host Country ambition, the Activity Developer shall apply the Downward Adjustment Factor (DAF) determined in Section 7.4. For Removal activities, "Adjusting Downwards" means increasing the Baseline Removals.

- 7.9.2. | **Calculation of Final Crediting Baseline ($BR_{crediting,y}$):**

$$BR_{crediting,y} = MAX([BR_{unadj,y} \times (1 + DAF_{total}) \times (1 + UNC_{BSL})], BR_{gov,y}) \quad (\text{Eq-6})$$

Where

Parameter	Description	Units
$BR_{crediting,y}$	Final Baseline Removals to be deducted from Activity Removals	tCO ₂ e
$BR_{unadj,y}$	Unadjusted Baseline Removals (from equation 1)	tCO ₂ e
DAF	The Downward Adjustment Factor (Fixed at 0.0125 for LUF)	Fraction
UNC_{BSL}	Baseline uncertainty factor (SE of match or 0.10 default)	Fraction
$BR_{gov,y}$	Removals mandated by government targets (if applicable).	tCO ₂ e

7.9.3. | **Track 3 Exception (Static Zero):** In cases where $BR_{unadj,y} = 0$ (e.g., Track 3 Revegetation or Reference Area Degradation), the mathematical adjustment above yields zero. In this case, the DAF Tool, the DAF shall instead be applied as a discount to the Activity Removals (AR_y) in Section 9.

8 | ACTIVITY SCENARIO

8.1 | Identification of Activity Emission Sources

8.1.1. | The Activity Scenario includes the implementation and management of the project activity (e.g., site preparation, planting, fertilization) that results in GHG emissions. The activity developer shall identify and account for Activity Emissions (AE_y) based on the assigned Compliance Track.

Table 13: Activity Emission Source Inclusion Matrix

Emission Source	Gas	Track 1 (Commercial / Large)	Track 2 & 3 (Smallholder / Restoration)	Justification
Biomass Burning (Site Prep / Wildfire)	CH ₄ , N ₂ O	Mandatory	Mandatory	Non-CO ₂ emissions from fire are significant regardless of scale and shall be accounted for.
Inorganic Fertilizer (Tree Establishment)	N ₂ O	Mandatory	Conditional <i>Excluded if application rate is < 50 kg N/ha/yr.</i>	Smallholder agroforestry typically relies on organic inputs or negligible synthetic quantities.

Fossil Fuel Use (Machinery / Transport)	CO ₂	Mandatory	Optional <i>Assumed De Minimis if manual labour is dominant.</i>	Fuel use in smallholder systems (motorbikes/hand-tools) is negligible compared to removal volumes.
Livestock (Silvopasture Only)	CH ₄	Mandatory	Mandatory	Only accounted if number of head of ruminants increases within the boundary compared to the baseline.

8.2 | Calculation of Total Activity Emissions

8.2.1. | The total Activity emissions (AE_y) for year y are calculated as the sum of emissions from all eligible sources defined in table 14.

$$AE_y = AE_{burn,y} + AE_{fert,y} + AE_{fuel,y} + AE_{livestock,y} \quad (\text{Eq-7})$$

Where

Parameter	Description	Units
AE_y	Total Activity Emissions in year y	tCO ₂ e
$AE_{burn,y}$	Emissions from biomass burning	tCO ₂ e
$AE_{fert,y}$	Emissions from synthetic fertilizer application	tCO ₂ e
$AE_{fuel,y}$	Emissions from fossil fuel consumption	tCO ₂ e
$AE_{livestock,y}$	Emissions from enteric fermentation/manure	tCO ₂ e

8.2.2. | **Simplified Option for Tracks 2 & 3 (Smallholders):** The activity developer may elect to apply a Conservative Default Deduction instead of monitoring specific fuel and fertilizer usage logs.

- Default Deduction:** 5% of Gross Activity Removals.
- Condition:** This option is permitted only if the activity developer provides evidence (e.g., survey data or attestation) that no heavy machinery (e.g., bulldozers) is used for site preparation.
- Application:** If elected,

$$AE_{fert,y} + AE_{fuel,y} = 0.05 \times AR_y \quad (\text{Eq-8})$$

($AE_{burn,y}$ and $AE_{livestock,y}$ shall still be monitored directly if they occur).

8.2.3. | **Standard Calculation (Mandatory for Track 1)**

8.2.3.1 | **Biomass Burning** ($AE_{burn,y}$) calculated if fire is used for site preparation or if wildfires occur within the boundary. CO₂ emissions are accounted for as a loss of carbon stock. Only non-CO₂ gases are calculated here.

$$AE_{burn,y} = \sum (M_{burn,i} \times EF_{gas} \times GWP_{gas}) \quad (Eq-9)$$

Where

Parameter	Description	Units
$M_{burn,i}$	Mass of dry biomass burned (tonnes d.m.)	tonnes d.m
EF_{gas}	Emission Factors for CH ₄ and N ₂ O (IPCC 2006 Vol 4, Ch 2, Table 2.5).	t gas/t d.m.
GWP_{gas}	Global Warming Potential of the specific gas	Dimensionless

8.2.3.2 | **Synthetic Fertilizer**

$$AE_{fert,y} = \sum (M_{N,i} \times EF_N \times \frac{44}{28} \times GWP_{N_2O}) \quad (Eq-10)$$

Where

Parameter	Description	Units
$M_{N,i}$	Mass of synthetic nitrogen fertilizer applied in area <i>i</i> (tonnes N).	tonnes N
EF_N	Emission factor for N ₂ O emissions from N inputs (kg N ₂ O-N / kg N input).	t N ₂ O-N / t N
$\frac{44}{28}$	Stoichiometric ratio converting N ₂ O-N to N ₂ O.	Dimensionless
GWP_{N_2O}	Global Warming Potential of N ₂ O.	Dimensionless

8.2.3.3 | **Fossil Fuel**

$$AE_{fert,y} = \sum (Vol_{fuel,i} \times EF_{CO_2,i}) \quad (Eq-11)$$

Where

Parameter	Description	Units
$Vol_{fuel,i}$	Volume or mass of fossil fuel type <i>i</i> consumed	liters or kg
$EF_{CO_2,i}$	Emission factor for fuel type <i>i</i> (e.g., Diesel = 2.68 kg CO ₂ /liter).	kgCO ₂ / liter (or kg)

9 | ACTIVITY REMOVALS

9.1 | Mechanism of Removal

9.1.1.1. | Activity Removals (AR_y) represent the Gross GHG Removals resulting from the accumulation of carbon in eligible pools (Biomass and Soil) within the Activity Boundary.

9.2 | Quantification Approach

9.2.1.1. | The Activity Developer shall use the approach corresponding to their Compliance Track (Section 3).

Table 14: Quantification Method Selection

Compliance Track	Method Name	Description	Best Used For
Track 1 (large scale)	Method A: Area-Based (Stratified Sampling)	$Area \times Density$ Uses permanent sample plots to estimate tC/ha.	Contiguous plantations or large-scale restoration.
Track 2 (Small and micro-scale)	Method A: Area-Based, or Method B: Census-Based (Digital Tree Count)	See above $Count \times Stock_{tres}$ Uses a count of individual trees (N).	Agroforestry, Dispersed Planting, Boundary Planting.
Track 3 (Revegetation, Community-led smallholder)	Method C: Allometric (Shrub/Harvest)	$Area \times Shrub_{vol}$ Uses canopy volume or destructive sampling.	Bamboo, Shrubs, Grasses.

9.3 | Calculation of Activity Removals ($AR_{gross,y}$)

9.3.1.1. | The Gross Activity Removals are calculated using the Stock Change Method across all strata and eligible pools:

$$AR_{gross,y} = \sum_i \left((C_{total,t2} - C_{total,t1}) \times \frac{44}{12} \right) \quad (\text{Eq-12})$$

Where

Parameter	Description	Units
$AR_{gross,y}$	Gross Activity Removals in year y (tCO ₂ e).	tCO ₂ e
i	Stratum index.	Dimensionless

$C_{total,t2}$	Total carbon stock in stratum i at the end of the monitoring period (tC).	tC
$C_{total,t1}$	Total carbon stock in stratum i at the beginning of the monitoring period (tC).	tC
$\frac{44}{12}$	Conversion factor from Carbon (C) to Carbon Dioxide (CO ₂).	Fraction

Note The Downward Adjustment Factor is applied as a discount ONLY for Track 3; see Step 5 below.

9.4 | Step-by-Step Calculation

9.4.1. | Step 1: Biomass Inventory (Measurement):

9.4.1.1 | For Method A (Area-Based - Track 1 or track 2):

- Method:** Establish Permanent Sample Plots (PSP) using Stratified Random Sampling.
- Measurement:** For all trees in the sample plot, measure tree parameters required by the allometric equation being used. These are usually diameter at Breast Height (DBH) and Height (H)
- Scaling:** Calculate mean biomass density per hectare (t d.m./ha) and multiply by the Stratum Area (A_i).

9.4.1.2 | For Method B (Census-Based - Track 2):

- Method:** Conduct a Digital Census of surviving trees. To ensure integrity, the census shall be supported by a preferably Digital MRV System (e.g., mobile app) capable of logging unique Tree IDs and GPS coordinates.
- Measurement:**
 - Count (N):** Total number of surviving trees recorded in the Digital MRV system.
 - Growth Parameters:** Measure DBH and Height for a Representative Sample (e.g., $n=30$ trees per cohort) to determine the "Mean Tree" size.
 - Estimate the mean tree biomass using the mean tree size and appropriate allometric equation.
- Scaling:** Multiply "Mean Tree Biomass" by the Census Count (N).

9.4.1.3 | For Method C (Track 3):

Use species-specific allometric models (e.g., Culm counts for Bamboo) or based on existing studies that are derived from destructive harvesting of quadrats to establish biomass density.

9.4.2. | Step 2: Calculate Biomass (Allometry):

9.4.2.1 | Convert field measurements into Aboveground Biomass (*AGB*) using peer-reviewed Allometric Equations.

9.4.2.2 | Allometric Equations

- a. Hierarchy: Species-specific (Local) > Species-specific (Regional) > Pantropical (e.g., Chave et al., 2014).
- b. Constraint: Wood Density (ρ) shall be specific to the species (e.g., sourced from the Global Wood Density Database).

9.4.3. | **Step 3: Belowground Biomass (*BGB*)**

9.4.3.1 | Unless destructively sampled, *BGB* (roots) is calculated using a Root-to-Shoot Ratio (*R*).

$$BGB = AGB \times R \quad (\text{Eq-13})$$

Where

Parameter	Description	Units
<i>BGB</i>	Belowground Biomass	t d.m. / ha
<i>AGB</i>	Aboveground Biomass	t d.m. / ha
<i>R</i>	Root-to-Shoot Ratio (IPCC defaults)	Dimensionless

Source: IPCC 2019 Refinement, Vol 4, Ch 4, Table 4.4 (e.g., $R \approx 0.20 - 0.30$ for tropical forests; $R \approx 0.40$ for shrublands).

9.4.4. | **Step 4: Soil Organic Carbon (*SOC*) [Optional]**

9.4.4.1 | Inclusion of *SOC* is **Optional**.

- a. **Method:** Measure ΔSOC via laboratory analysis of soil cores (0-50cm) at fixed intervals (min. 5 years).
- b. **Discount:** Due to high measurement variability, a 20% discount is applied to the measured SOC increase unless statistical significance ($p < 0.05$) is demonstrated.

Question for Stakeholders:

The proposed methodology utilizes field-based sampling and direct measurement of soil organic carbon (SOC) to quantify sequestration enhancements. Consequently, it proposes retiring the existing SOC tool currently permitted under the GS forestry methodology.

Feedback: Stakeholders are invited to comment on whether the methodology should allow for the use of default values in specific scenarios.

Feedback Requested: Stakeholders are invited to comment on whether the methodology should allow for the use of default values in specific scenarios. For instance, could a conservative default value that is specific to the project activity area; be considered if it is derived from peer-reviewed literature that includes comprehensive data on variables and margins of error? What are the guardrails to be considered if default values are allowed for SOC quantification.

9.4.5. | Step 5: Uncertainty and DAF Application**9.4.5.1 | Uncertainty Deduction (UNC_{AR}):**

- a. Calculate the aggregate uncertainty of the biomass estimate at the 90% Confidence Interval (CI).
 - i. If Uncertainty $\leq 10\%$: No deduction.
 - ii. If Uncertainty $\geq 10\%$: Deduct the excess percentage

$$AR_{adjusted,y} = AR_{gross,y} \times (1 - (UNC_{Total} - 10\%)) \quad (\text{Eq-14})$$

Where

Parameter	Description	Units
$AR_{adjusted,y}$	Activity Removals adjusted for uncertainty	tCO ₂ e
$AR_{gross,y}$	Gross Activity Removals	tCO ₂ e
UNC_{Total}	Total combined uncertainty of the biomass estimate	Percentage (%)

9.4.5.2 | Track 3 DAF Discount (Special Provision): As stipulated in Section earlier, since Track 3 baselines are set to Zero, the Downward Adjustment Factor (DAF) determined in Section 7 cannot be applied to the baseline. Instead, it is applied here as a discount to the removals.

- a. For Track 3 Only:

$$AR_{final,y} = AR_{adjusted} \times (1 - DAF) \quad (\text{Eq-15})$$

- b. For Track 1 & 2:

$$AR_{final,y} = AR_{adjusted,y} \quad (\text{Eq-16})$$

Where

Parameter	Description	Units
$AR_{final,y}$	Final Activity Removals to be credited	tCO _{2e}
DAF	The Absolute Downward Adjustment Factor (0.0125)	Fraction

10| LEAKAGE EMISSIONS

10.1 | Identification of Leakage Sources (Step 1)

- 10.1.1. | In accordance with Section 6.2 of the GS4GG PAA MS400-05, the activity developer shall identify all potential sources of leakage emissions—defined as the net increase in anthropogenic GHG emissions occurring outside the Activity Boundary that is attributable to the implementation of the project activity.
- 10.1.2. | The applicability of specific leakage sources is determined by the Activity Module assigned to the project in Section 3. The activity developer shall account for the sources indicated in Table 15.

Table 15: Leakage Source Identification Matrix

Leakage Category	Activity Shifting (LE_{shift})	Market Effects (LE_{market})	Ecological (LE_{eco})
Description	Displacement of pre-project agricultural, grazing, or fuel-collection activities to areas outside the boundary.	Emissions triggered by a reduction in the supply of goods (e.g., timber, crops) causing production shifts outside the leakage buffer area.	Emissions resulting from hydrological displacement (e.g., wetland desiccation downstream).
Module 1	Mandatory (Spatial Monitoring)	Conditional (If productive land converted)	Excluded (De Minimis)
Module 2	Mandatory (Spatial Monitoring)	Excluded	Excluded
Module 3	Included	Conditional (Yield Test)	Excluded
Module 4	Included	Excluded	Excluded
Module 5	Excluded	Excluded	Excluded

10.1.3. | Justification for Exclusions

- 10.1.3.1 | **Module 5 (Revegetation):** Activity shifting is excluded because eligibility requires the land to be degraded or abandoned (supporting no productive economic activity) for >3 years prior to the start date.

10.1.3.2 | **Ecological Leakage:** Excluded for all modules as the Applicability Conditions (Section 3) strictly prohibit activities on wetlands or areas where hydrological diversion affects downstream carbon pools.

10.2 | Avoidance and Minimization Strategies (Step 2)

10.2.1. | In accordance with Section 6.3 of *GS4GG PAA MS400-05*, the Activity Developer shall prioritize the avoidance of leakage over ex-post quantification. The Project Design Document (PDD) shall detail the specific strategies employed to prevent displacement, corresponding to the Activity Module:

10.2.1.1 | **Supply Chain Integration (Module 1 and Module 3):** The activity shall prioritize the use of degraded or marginal lands rather than converting productive agricultural land. Where conversion occurs, the developer shall demonstrate that intensification elsewhere offsets the lost production.

10.2.1.2 | **Livelihood Diversification (Modules 2, 3, 4 and 5):** Where the baseline activity involves subsistence grazing or charcoal production, the activity shall include a Livelihood Action Plan to provide alternative income streams (e.g., beekeeping, NTFP harvesting) or sustainable fodder banks to reduce pressure on adjacent forests.

10.2.1.3 | **Sustainable Intensification (Modules 3 and 4):** For agroforestry activities, the design shall ensure that tree planting does not reduce the total food production of the farm. Strategies shall include intercropping, the use of nitrogen-fixing species, or improved agronomic inputs.

10.3 | Quantification of Leakage Emissions (Step 3)

10.3.1. | Where leakage cannot be fully avoided, the Activity Developer shall quantify and deduct leakage emissions (LE_y) from the Net verified removals. The quantification method is strictly defined by the Compliance Track.

10.3.2. | **Activity Shifting Leakage ($LE_{shift,y}$)**

10.3.2.1 | **Method A: Leakage Belt Monitoring (Track 1).**

a. **Applicable to:** Module 1 (Commercial forestry) and Module 2 (Restorative Forestry).

b. **Belt Establishment:** The Activity Developer shall delineate a spatial **Leakage Belt** surrounding the Activity Boundary.

a. **Width:** Minimum 2 km (Agricultural displacement) to 5 km (Grazing displacement).

b. **Constraint:** The belt shall consist of lands suitable for the displaced activity (e.g., excluding water bodies or settlements).

c. **Monitoring:** During each monitoring period y , the Developer shall monitor the Leakage Belt for Forest Cover Loss ($Area_{def,y}$) using the same remote sensing protocols defined for the Baseline (Section 7).

Questions for Stakeholders:

The draft currently suggests leakage belts of 2 km and 5 km. However actual leakage may occur across larger landscapes or be difficult to attribute directly, often resulting in high monitoring costs that are disproportionate to the actual leakage volume.

Feedback Requested: We seek feedback on: (i) the relevance of a leakage belt-based monitoring in context of ARR activities (ii) the current 2 km/5 km approach; (iii) is there a more cost-effective or scientifically robust method for quantifying leakage that avoids the burden of extensive belt monitoring without compromising the accuracy of the net GHG benefits?

d. Calculation:

$$LE_{shift,y} = MAX(0, (Area_{def,y} - Area_{baseline,y}) \times C_{belt}) \times \frac{44}{12} \quad (\text{Eq-17})$$

Where

Parameter	Description	Units
$LE_{shift,y}$	Leakage emissions from activity shifting	tCO ₂ e
$Area_{def,y}$	Deforested area observed in the Leakage Belt in year y	ha
$Area_{baseline,y}$	Historical baseline deforestation area in the Leakage Belt	ha
C_{belt}	Conservative average carbon stock of vegetation in the belt	tC/ha

Where $Area_{def,y}$ is the historical deforestation rate in the belt and C_{belt} is the conservative carbon stock (tC/ha) of the vegetation at risk.

10.3.2.2 | Method B: Yield Maintenance Test (Track 2)

- Applicable to:** Modules 3 and 4 (Agroforestry). For displaced smallholders, spatial belts are impractical. Instead, a "Yield Maintenance Test" serves as a proxy for displacement.
- Test:** Monitor the yield of the primary subsistence crop (Y_{crop}) on a representative sample of participating farms.
- Trigger:** Compare Project Yield (Y_p) to Baseline Yield (Y_b).
 - Pass:** If $Y_p \geq 0.90 \times Y_b$ (Yield decline <10%), then $LE_{shift,y} = 0$
 - Fail:** If yield declines by >10%, displacement is assumed.
- Deduction (If Fail):** A flat penalty applies to the activity removals (AR_y) of the failing cohort to account for the land required to replace the lost food.

$$LE_{shift,y} = 0.15 \times AR_y \quad (\text{Eq-18})$$

Where

Parameter	Description	Units
$LE_{shift,y}$	Leakage emissions from activity shifting (Yield test fail)	tCO ₂ e
AR_y	Activity Removals of the failing cohort	ha
0.15	Default conservative penalty factor (15%)	dimensionless

10.3.2.3 | Method C: Deemed Zero (Track 3)

- Applicable to: Modules 5 (Revegetation) Sub-Module A (Smallholder Aggregation).**
- Leakage is **Deemed Zero** provided the activity developer demonstrates (via historical imagery or participatory rural appraisal) that the land was degraded/abandoned for at least **3 years** prior to the project start date.

10.3.3. | Market Leakage ($LE_{market,y}$)

- Applicable to:** Modules 1 and 3 (Commercial) only.
- Trigger:** Required if the activity displaces Commercial Commodity Agriculture (e.g., converting Soy or Cattle lands to Timber) leading to a supply deficit.

$$LE_{market,y} = AR_y \times LF_{market} \quad (\text{Eq-19})$$

Where

Parameter	Description	Units
$LE_{market,y}$	Leakage emissions from activity shifting (Yield test fail)	tCO ₂ e
AR_y	Gross Activity Removals	tCO ₂ e
LF_{market}	Leakage Factor (0.00, 0.05, or 0.10)	Fraction

- $LE_{market,y}$ (Leakage Factor):
 - (0%): If the activity occurs on degraded land (No displacement).
 - 0.05 (5%): If displacing crops grown for domestic/subsistence markets.
 - (10%): If displacing globally traded commodities (e.g., Soy).

10.3.4. | Total Leakage Calculation

10.3.4.1 | The Total Leakage Emissions (LE_y) to be deducted from the Net Removals are calculated as:

$$LE_y = LE_{shift,y} + LE_{market,y} \quad (\text{Eq-20})$$

- 10.3.5. | **Uncertainty Application (UNC_{LE}):** In accordance with GS4GG requirements, if the leakage calculation relies on default factors (e.g., Method B Yield Deduction or Market Leakage Factors) rather than direct measurement, a conservative uncertainty buffer shall be added.

$$LE_{final,y} = LE_y + (1 + 0.15) \quad (\text{Eq-21})$$

Where

Parameter	Description	Units
$LE_{final,y}$	Final Leakage Emissions (including 15% uncertainty buffer)	tCO ₂ e
LE_y	Total unadjusted leakage emissions	tCO ₂ e
0.15	Uncertainty buffer for reliance on default leakage factors	dimensionless

11 | CALCULATION OF NET GHG REMOVALS

11.1 | Net Quantification Approach

- 11.1.1. | The activity developer shall calculate the net verified removals for the monitoring period y . This metric represents the total quantity of Carbon Dioxide Equivalent (tCO₂e) removed from the atmosphere and durably stored by the project activity, net of all deductions for baseline removals, activity emissions, leakage, and uncertainty.

11.2 | Calculation of Net Verified Removals (nAR_y)

- 11.2.1. | The Activity Developer shall calculate nAR_y using the following equation:

$$nAR_y = (AR_{final,y} - AE_y) - (BR_{crediting,y} - BE_{MU,y}) - LE_{final,y} \quad (\text{Eq-22})$$

Where

Parameter	Description	Units
nAR_y	Net Verified Removals in year y	tCO ₂ e
$AR_{final,y}$	Activity Removals (Final) (tCO ₂ e): Gross biomass growth in the project area, adjusted for uncertainty. <i>Note: For Track 3, this value includes the DAF Discount applied in Section 9.4.(Section 9)</i>	tCO ₂ e
AE_y	Activity Emissions (tCO ₂ e): GHG emissions from project operations (fuel, fertilizer, biomass burning). (Section 8)	tCO ₂ e
$BR_{crediting,y}$	Crediting Baseline Removals (tCO ₂ e): The deduction for natural regeneration. <i>Note: For</i>	tCO ₂ e

	Tracks 1 & 2 , this value includes the DAF Increase applied in Section 7. (Section 7)	
BE_y	Baseline Emissions (tCO ₂ e): Set to Zero (0) for conservativeness (Avoided emissions are not credited). (Section 7)	tCO ₂ e
$LE_{final,y}$	Leakage Emissions (tCO ₂ e): Deduction for activity shifting or market effects, including uncertainty buffers. (Section 10)	tCO ₂ e

11.3 | Confirmation of Downward Adjustment (DAF) Application

11.3.1. | To ensure compliance with the GS4GG Methodology Standard, the activity developer shall confirm in the Monitoring Report that the Downward Adjustment Factor (DAF_{Total}) has been correctly applied based on the Compliance Track to prevent double counting or omission:

- Track 1 & 2: Confirmed that DAF was applied to increase the Baseline Removals ($BR_{crediting,y}$).
- Track 3: Confirmed that DAF was applied to discount the Activity Removals ($AR_{final,y}$).

11.4 | Negative Balance (Deficit Carryover)

11.4.1. | If the calculated nAR_y is Negative (i.e., Deductions + Emissions > Removals):

- Issuance: The Issuable GSVERs for the current period shall be Zero.
- Carryover: The absolute value of the negative balance shall be recorded as a Negative Carryover.
- Future Deduction: This deficit shall be fully offset against positive removals in subsequent monitoring periods (y+1) before any new credits can be issued.

11.5 | Determination of Buffer Rate

11.5.1. | The activity developer shall determine the applicable Risk Rating based on the Compliance Track.

Table 16: Buffer Contribution Rates

Compliance Track	Risk Profile	Buffer Rate	Conditions / Incentives
Commercial Forestry	High Risk	20% (Default)	Standard deduction for commercial forestry activities with harvesting risks.

Restorative Forestry	High Risk	20%	High risk of project failure due to lack of financial support, fire and other environmental risks
Integrated Land Use including stacking of Module 5 (Revegetation)	Medium Risk	10% - 20%	The rate may be reduced to 10% if the activity demonstrates that the trees provide essential livelihood benefits (e.g., fruit, fodder valued at >20% of household income), creating a rational economic disincentive for reversal.
Revegetation (no stacking)	Variable Risk	20% (Default)	Standard deduction due to the susceptibility of tree and non-tree vegetation (shrubs/grasses) to fire and grazing.
Smallholder aggregation	Medium Risk	10% - 20%	For the same reasons as integrated land use

11.6 | Final Issuable GSVERs

11.6.1. | The final number of Gold Standard Verified Emission Reductions (GSVERs) to be issued to the Activity Developer is calculated as:

$$\text{Issuable GSVERs} = nAR_y \times (1 - BP_y) \quad (\text{Eq-23})$$

12 | MEETING METHODOLOGICAL PRINCIPLES

12.1 | Encouraging ambition over time

12.1.1. | This methodology integrates dynamic mechanisms to ensure that baselines evolve dynamically, incentivizing progressively higher mitigation performance over the project lifetime:

- Dynamic Baseline Adjustment (SEMDB):** For **Track 1** activities, the use of **Spatially Explicit Matched Dynamic Baselines (SEMDB)** ensures that credits are only issued for removals strictly *above* real-time natural regeneration. If climate conditions (e.g., CO₂ fertilization or increased rainfall) accelerate natural background growth, the baseline deduction automatically rises.
- Downward Adjustment Factor (DAF):** As detailed in **Section 7.4**, the methodology applies a mandatory DAF derived from the Host

Country's Net Zero trajectory or Sectoral Targets. This factor progressively increases the baseline removals (for Tracks 1 & 2) or the removal discount (for Track 3) over the crediting period, ensuring that the activity remains additional to the evolving national decarbonization rate.

- c. **Baseline Validity Limits:** The "Locked Map" and "Regional Curve" parameters have a maximum validity of **10 years** (Section 7.3). Upon renewal, baselines shall be re-validated to reflect the latest socio-economic and climatic reality, ensuring the reference level never remains static in a changing world.

12.2 | Equitable sharing of mitigation benefits

12.2.1. | The methodology is explicitly designed to address access gap and ensure inclusivity for all stakeholders, particularly smallholders and local communities.

12.2.2. | **Modular Architecture:** By establishing a compliance Track approach, the methodology removes the barrier of high-cost, remote sensing for small and microscale projects. This allows community-led agroforestry and Revegetation with non-tree woody species to access carbon finance using standardized, accessible data (Regional Curves) without compromising rigor.

- a. **Social Permanence Incentive:** As defined in Section 13.2, activities that demonstrate tangible livelihood benefits (limited to Module 2, 3 or 4) (e.g., food security, income from sustainable harvesting) are eligible for a reduced Buffer Contribution Rate (10% as per section 11.5). This explicitly values the social co-benefits of the project as a driver of risk reduction and permanence.

Questions for Stakeholders:

Social Permanence and Community-Managed Systems

The methodology introduces the concept of **Social Permanence**. This recognizes that permanence can be naturally safeguarded when local communities rely on the long-term sustenance and non-timber forest products (NTFPs) provided by the trees. This is applicable to Module 2, 3 or 4.

Feedback Requested: We invite comments on this sustenance-based approach to permanence. Additionally, we are looking for alternative examples or metrics; beyond conventional legal or financial instruments that can effectively assess the long-term stability of plantations.

- b. **Benefit Sharing:** For activities aggregated under a Programme of Activities (PoA), the activity developer shall demonstrate transparent benefit-sharing arrangements, ensuring that carbon revenues flow to the

smallholders managing the trees, consistent with the GS4GG Land Use & Forests Activity Requirements.

12.3 | Avoidance of double counting

12.3.1. | The methodology employs robust safeguards to prevent double issuance, double use, and double claiming of mitigation outcomes.

- a. **Spatial Exclusivity:** The specific geodetic delineation of the Activity Planting Area and the Dynamic Reference Area (Section 5.1) ensures that no two projects can claim the same land or use the same control plots for conflicting baselines.
- b. **Registry Tagging:** All GSVERs issued under this methodology are tagged as "Removals" in the Gold Standard Registry. The methodology strictly differentiates between removals (credited) and avoided Emissions (excluded in Section 7.8), preventing attribute conflation.
- c. **Regulatory Surplus:** The mandatory Regulatory Surplus Analysis (Section 6.3) ensures that the activity is not claiming credits for removals that are already legally mandated or accounted for under specific government compliance schemes (outside of the NDC).

12.4 | Aligning with NDC and LT-LEDs

12.4.1. | The methodology ensures that activities support, rather than displace, the Host Country's Nationally Determined Contributions (NDC) and Long-Term Low-Emission Development Strategies (LT-LEDs).

- a. **The Government Target Floor:** As mandated in Section 7.9, the Crediting Baseline ($BR_{crediting,y}$) is capped at the level of any unconditional NDC reforestation target or legal replanting mandate. This ensures that carbon finance is not used to subsidize activities that the government has already committed to achieving.
- b. **Sectoral Alignment:** The DAF mechanism prioritizes Host Country-specific Sectoral Factors (where available) over global defaults, ensuring the project aligns with the specific decarbonization curve of the national LULUCF sector.

12.5 | Encouraging broad participation

12.5.1. | The methodology moves away from a "one-size-fits-all" approach to encourage participation across diverse scales and project types.

- a. **Routing Matrix:** Section 3 establishes a clear decision framework that routes large scale projects to high-rigor monitoring while small and micro-scale projects to simplified, appropriate quantification methods. This ensures that stringent requirements for industrial actors do not disenfranchise smallholder developers.

- b. **Deemed Additionality:** The use of Positive Lists for LDCs, SIDS, and non-commercial restoration projects (Section 6.2) reduces the administrative burden of financial auditing for the projects that need financing the most.
- c. **Census-Based Monitoring:** The inclusion of Method B (Census-Based) quantification (Section 8.2) allows farmers to participate by counting trees via digital apps rather than establishing complex permanent forestry plots.

12.6 | Including data sources, accounting for uncertainty and monitoring

12.6.1. | The methodology ensures robustness in data sourcing, monitoring, and uncertainty management through the following provisions:

- a. **Tiered Data Hierarchy:** The methodology prioritizes high-quality local data (Method A/Track 1) but permits the use of robust regional aggregates (Method B/Track 2) where local data is unavailable, balancing accuracy with feasibility.
- b. **Integrated Uncertainty Accounting:** Uncertainty is not treated as an afterthought but is integrated into every calculation step.
 - i. Baseline: Track 1 calculates statistical uncertainty (SE) of matched pixels; Tracks 2 & 3 apply default deductions (10-20%) in Section 7.4.
 - ii. Removals: Section 9.4 applies a deduction if biomass confidence intervals exceed 10%.
 - iii. Leakage: Section 10.3 applies a 15% buffer to default leakage factors.
- a. **Digital Verification:** For distributed activities (Track 2), the methodology mandates the use of digital data collection tools (e.g., GPS-tagged photos, mobile logs) to ensure auditability of the "Census" count.

12.7 | Taking into account policies, measures and relevant circumstances

- a. **Leakage Management:** The leakage provisions (Section 10) explicitly account for local market dynamics and land-use pressure. For example, the Yield Maintenance Test (Section 11.3.2) replaces abstract modelling with a practical check on food security, recognizing that in subsistence contexts, food production is the primary driver of land-use change.
- b. **Legal Compliance:** The activity shall demonstrate compliance with all relevant national and sub-national laws, including land tenure regulations. The "Locked Map" protocol includes a mandatory audit for Anthropogenic Land-Use Change ([Annex 1](#)), ensuring that the baseline

reflects real-world policy outcomes (e.g., zoning changes) rather than hypothetical scenarios.

13| REVERSALS

13.1 | Definitions and Scope

- 13.1.1. | A Reversal is defined as a verified decrease in the carbon stocks of the eligible pools (biomass and soil) within the Activity Boundary below the levels credited in previous monitoring periods. A reversal signifies a release of previously sequestered carbon back into the atmosphere.
- 13.1.2. | For the purpose of attribution and compensation, reversals are classified into two categories
- a. **Unavoidable Reversal (Force Majeure):** A reversal caused by natural events beyond the reasonable control of the Activity Developer (e.g., wildfire, extreme weather, pest outbreak, or geological disturbance), provided that mandatory risk mitigation measures were implemented.
 - b. **Avoidable Reversal:** A reversal caused by human activities (e.g., harvesting, land clearance, negligence) or the failure of the Activity Developer to implement the required risk mitigation plan.
- 13.1.3. | Gold Standard Risks and Capacities assessment shall be conducted to identify reversal risks and prepare risk mitigation plans to minimize risk reversals.

13.2 | Assessment of Reversal Risks

- 13.2.1. | The activity developer shall manage non-permanence risk through the Gold Standard Compliance Buffer. As quantified in Section 11.5, a portion of the Net Verified Removals is contributed to the buffer at each issuance event.
- 13.2.2. | **Social Permanence (applicable to Integrated Land Use and Smallholder Aggregate activities only)**
- 13.2.3. | For Integrated Land Use for subsistence (Module 4) and Smallholder aggregator (integrated into Module 2, 3 or 4) activities utilizing Track 2, the methodology recognizes "Social Permanence" as a primary risk mitigation tool.
- a. **Principle:** Unlike commercial forestry projects, trees integrated into farming systems provide essential livelihood benefits (e.g., fruit, fodder, shade for crops), creating a rational economic incentive for the participant to retain the tree ("The tree is worth more alive than dead").
 - b. **Requirement:** To qualify for the reduced Buffer Rate (10%) defined in Section 11.5, the Activity Developer shall demonstrate in the PDD that the "Livelihood Benefit Ratio" (Value of Tree Services / Total Farm Income) is >20% for the participating cohort.

13.3 | Risk Mitigation Measures

13.3.1. | The activity developer shall implement the following mandatory risk mitigation measures based on the **Activity Module**:

Table 17: Mandatory Risk Mitigation Measures [EXAMPLES ONLY]

Activity Module	Primary Risks	Mandatory Mitigation Measure
M1: Commercial Forestry	Harvest / Fire	Fire Breaks: Establishment of fire breaks based on professional assessment of the land, and wind patterns. Legal Protection: Registration of a restrictive covenant or long-term lease prohibiting land-use change.
M2: Restorative Forestry	Fire / Encroachment	Fire Breaks: See above Community Patrols: Implementation of a community-led guarding program. Benefit Sharing: Transparent distribution of carbon revenues to surrounding communities to reduce encroachment pressure.
M3 and M4: Integrated Land Use	Land Turnover	Contractual Tenure: Agreements shall clarify that carbon rights (and survival obligations) transfer with the land title if the farm is sold. Diversification: Planting of mixed species (min. 3 species) to prevent pest-induced catastrophic failure.
M5: Revegetation	Grazing / Fire	Living Fences: Establishment of vegetative barriers to exclude livestock. Fuel Management: Active management of dry biomass loads (e.g., controlled early burning) to prevent high-intensity wildfires.

13.4 | Monitoring for Reversals

13.4.1. | The activity developer shall monitor the project area for reversals throughout the Crediting Period. The monitoring protocol is determined by the Compliance Track.

13.4.1.1 | Track 1: Remote Sensing (Satellite Alerts)

- a. **Applicability:** Large Scale Projects.
- b. The activity developer shall subscribe to a near-real-time deforestation alert system (e.g., Global Forest Watch GLAD-L or RADD alerts).

- c. If disturbance alerts cover >5% of the Activity Planting Area in any reporting year, a Verification Event is triggered immediately to quantify the loss.

13.4.1.2 | Track 2: "Representative Project Farm"

- a. **Applicability:** Integrated Land Use or smallholder aggregates.
- b. **Representative Project Cohort:** Conducting a census of thousands of land owners annually is cost prohibitive. This track uses a statistical sample to detect reversals. The activity developer shall randomly select 1% of participating land owners (or a minimum of 30 land owners) annually to serve as "Representative Project Cohort."
- c. **Monitoring:** These lands are physically inspected (via app-based Census) to calculate a Disturbance Rate.
 - i. If Disturbance Rate is stable ($\pm 5\%$ of previous verification): No action required.
 - ii. If Disturbance Rate increases by >10%: A full stratified census of the entire project area is triggered to quantify the reversal.

13.5 | Quantification of Reversals (REV_y)

13.5.1. | In the event of a confirmed reversal, the Activity Developer shall quantify the total loss of carbon stocks (tCO_{2e}). This may be done using two methods: area-based percentage loss or a statistical monitor of the biomass remaining on the disturbed lands

13.5.1.1 | **Method A: Area-based percentage loss:** This method assumes that all biomass on the disturbed lands is lost

$$REV_y = \left(C_{verified,t-1} \times \frac{Disturbed\ Area}{Total\ Area} \times \frac{44}{12} \right) + AE_{burn,y} \quad (Eq-24)$$

13.5.1.2 | **Method B:** Monitoring of Remaining Biomass

$$REV_y = \left((C_{verified,t-1} - C_{monitored,t}) \times \frac{44}{12} \right) + AE_{burn,y} \quad (Eq-25)$$

Where

Parameter	Description	Units
REV_y	Total reversal amount.	tCO _{2e}
$C_{verified,t-1}$	Total carbon stock verified in the previous monitoring period	tC
$C_{monitored,t}$	Remaining carbon stock measured immediately following the reversal event.	tC
<i>Disturbed Area</i>	Area impacted by the reversal event.	ha
<i>Total Area</i>	Total Activity Area prior to the reversal.	ha

$AE_{burn,y}$	Non-CO ₂ emissions calculated per Section 8.2 (if reversal caused by fire)	tCO ₂ e
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13.6 | Addressing Reversals (Buffer Approach)

13.6.1. | **Compensation for Unavoidable Reversals:** If the VVB confirms the reversal was unavoidable (Force Majeure):

- a. **Buffer Drawdown:** Gold Standard shall cancel an equivalent number of credits from the Global Compliance Buffer Pool.
- b. **No Liability:** The activity developer is not financially liable for replenishing the buffer, provided risk mitigation measures were active.
- c. **Baseline Reset:** The project crediting calculation resumes from the new, lower stock level.

13.6.2. | **Compensation for Avoidable Reversals:** If the VVB determines the reversal was **Avoidable** (e.g., harvest without replanting, negligence):

- a. **Developer Liability:** The activity developer shall retire an equivalent number of GSVERs (or eligible compliance units) to compensate for the reversed volume.
- b. **Buffer Prohibition:** The Buffer Pool cannot be used to cover avoidable reversals.
- c. **Suspension:** Issuance of new credits is suspended until the reversal is fully compensated and a remediation plan is approved.

13.7 | Monitoring for Reversals and Liability – Post crediting period

13.7.1. | The activity developer is responsible for monitoring the project area to identify any potential reversal risks or events throughout the crediting period, and for the duration specified in the latest version of the Land Use & Forests Activity Requirements.

13.7.2. | A post-crediting period permanence mechanism shall be built into project design in the ARR project activity. This may include but not limited to utilizing tools such as conservation easements, protected land and other effective conservation land that commercial harvesting and forest conversion are not allowed, formalizing land tenure for local rightsholders, and designing project

13.7.3. | activities likely to lead to long term behaviour change up to 100 years.

Question for Stakeholders:

The methodology proposes a crediting period of 40-100 years. It also proposes post crediting monitoring, where projects are required to consider post crediting monitoring in the activity design.

Feedback: We seek feedback and suggestions on post crediting monitoring requirements.

Question 1: How should the post crediting monitoring period be determined? Is a formula, based on activity specific factors better, or is a uniform 100-year requirement preferred?

Question 2: What criteria or variables (e.g., reversal risk, ecosystem type, disturbance likelihood) should be considered if a formula-based approach is preferred to fix the years to be considered for post crediting monitoring?

Question 3: What challenges (e.g., financial, technical, or institutional) would activity developers face in meeting long term monitoring obligations beyond the crediting period?

Question 4: What types of incentives (financial or non-financial) would meaningfully encourage developers to continue monitoring after credits are no longer issued?

Question 5: How should the methodology ensure continuity of monitoring if activity ownership changes or the original developer is no longer active?

14| UNCERTAINTY QUANTIFICATION

14.1 | Approach for Uncertainty Management

- 14.1.1.| The activity developer shall account for uncertainty in the quantification of GHG removals and emissions to ensure that the Net Verified Removals (nAR_y) are conservative.
- 14.1.2.| In accordance with the Sampling and Estimation of Parameters, all sampling estimates shall achieve a precision of $\pm 10\%$ of the mean at the 90% Confidence Interval (CI).
- 14.1.3.| To accommodate the Modular Architecture, this methodology manages uncertainty at the Component Level. This ensures that high-precision components (e.g., a 100% Tree Census) are not unfairly penalized by the uncertainty of lower-precision components (e.g., Leakage defaults).

14.2 | Summary of Uncertainty Requirements

- 14.2.1.| The activity developer shall identify and quantify the following sources of uncertainty and apply the deductions as specified in Table the table below.

Table 18: Uncertainty Application Matrix

Component	Source of Error	Method / Requirement
Baseline (UNC_{BSL})	Large Scale Projects: Standard Error (SE) of the Synthetic Control matches. Small and microscale projects: Default Uncertainty factor (10-20%).	Added to the Downward Adjustment Factor (DAF) to increase the baseline deduction.
Removals (UNC_{AR})	Large Scale Projects: Sampling error in change of biomass. Small and micro scale: Sampling error of "Mean Tree" biomass. Revegetation and Smallholder aggregate activities: Allometric model error.	Deducted from Gross Activity Removals if Total Uncertainty >10%.
Leakage (UNC_{LE})	Reliance on default leakage factors (e.g., Yield Deduction) rather than direct spatial monitoring.	Added to Leakage Emissions to increase the deduction.

14.3 | Calculation of Removal Uncertainty (UNC_{AR})

14.3.1. | The activity developer shall calculate the Total Uncertainty for Activity Removals (UNC_{AR}) using the method corresponding to the assigned **Compliance Track**.

14.3.2. | **Large Scale Projects: Area-Based (Stratified Sampling)**: For projects using Permanent Sample Plots (PSP), uncertainty is driven by the spatial variance of change in biomass density (tC/ha).

$$UNC_{AR} = \frac{t_{val} \times SE_{weighted}}{Mean_{weighted}} \times 100 \quad (\text{Eq-26})$$

Where

Parameter	Description	Units
UNC_{AR}	Sampling uncertainty of Activity Removals	Percentage (%)
t_{val}	Two-sided Student's t-value for 90% confidence (n-1 degrees of freedom).	Dimensionless
$Mean_{weighted}$	Mean net biomass density change weighted across all strata.	tC/ha
$SE_{weighted}$	Standard Error of the mean net biomass change weighted across all strata.	tC/ha

14.3.3. | **Track 2: Census-Based (Hybrid Statistics):** For small scale projects, the "Population Size" (Tree Count, N) is known from the 100% Digital Census with effectively zero sampling error (subject to QA/QC). Uncertainty applies only to the "Mean Tree" estimate derived from Representative Area. *Note: Since N is fixed, the relative uncertainty of the total biomass is equal to the relative uncertainty of the "Mean Tree" biomass estimate.*

- a. **Calculation:** Calculate the Standard Error of the "Mean Tree" size from the Representative Area measurements and convert to percentage uncertainty.

14.3.4. | **Aggregate Model Error (All Tracks):**

14.3.4.1 | If the activity uses default Allometric Equations (rather than locally or regionally derived allometric equations based on destructive sampling), a default Model Uncertainty (U_{model}) of 10% shall be combined with the Sampling Uncertainty ($U_{sampling}$) using the Root Sum of Squares (RSS).

$$UNC_{Total} = \sqrt{U_{model}^2 + U_{sampling}^2} \quad (\text{Eq-27})$$

Where

Parameter	Description	Units
UNC_{Total}	Total aggregated uncertainty for the activity	Percentage (%)
U_{model}	Model uncertainty (Default 10% for uncalibrated allometry)	Percentage (%)
$U_{sampling}$	Sampling uncertainty (UNC_{AR} or representative census error)	Percentage (%)

14.3.4.2 | The allometric equation may be developed for the species within an agro-ecological region. If such allometric equations are applied with demonstrable justifications (comparable agro-ecological parameters etc. supported by scientific literature), such equations shall be considered adequate to be used for the species within the local agro-ecological conditions.

14.3.5. | **Deduction Application**

14.3.5.1 | The calculated UNC_{Total} determines the deduction to be applied:

- If $UNC_{Total} \leq 10\%$: No Deduction.
- If $UNC_{Total} > 10\%$: Deduct the excess percentage ($UNC_{Total} - 10\%$).

14.4 | Systematic Error Check (Calibration)

14.4.1. | To manage systematic error (bias), the following calibration checks are mandatory:

- a. Instrument Calibration: All physical measurement devices (e.g., DBH tapes, hypsometers, weigh scales) shall be calibrated annually against a known standard.
- b. Digital Logic Audit (Track 2): Prior to verification, the Digital MRV system algorithms (summing tree counts) shall be audited to ensure zero calculation error.
- c. Bias Correction: If the Internal Audit reveals a systematic deviation >5% between field teams and the auditor, the data shall be corrected by the mean deviation percentage to be conservative.

15| MONITORING METHODOLOGY

15.1 | General Requirements

- 15.1.1. | The activity developer shall establish and maintain a comprehensive Monitoring Plan to quantify the Net Verified Removals (nAR_y) generated by the activity. The monitoring procedures shall be operationalized prior to the start of the first verification event and maintained throughout the Crediting Period.
- 15.1.2. | **Monitoring Principles (by Track):** The monitoring system shall adhere to the following principles based on the assigned **Compliance Track** (Section 3):
 - a. **Large Scale Projects (Track 1):** Shall utilize Method A (Area-Based) monitoring. This relies on remote sensing for area delineation ($A_{project}$) and stratified Permanent Sample Plots (PSP) for biomass quantification.
 - b. **Small and Microscale Projects (Track 2):** Shall utilize Method B (Census-Based) monitoring. This relies on a Digital MRV System to conduct a 100% census of trees (N) supported by Representative Area sampling for growth parameters (DBH, H).
 - c. **Revegetation and Smallholder activities (Track 3):** Shall utilize Method C (Allometric) monitoring. This relies on volumetric sampling of tree and non-tree biomass.

15.2 | Monitoring Frequency

- 15.2.1. | Monitoring activities shall be conducted at the frequency specified below:
 - a. **Continuous:** Activity data (e.g., planting records, fertilizer use, disturbance alerts) shall be recorded continuously.
 - b. **Annual:** Reversal Screening: Check of satellite disturbance alerts (Track 1) or Representative Area survival checks (Track 2).
- 15.2.2. | Periodic (Verification):
 - a. Full biomass inventory and reporting shall occur at least every 5 years (mandatory).

- b. **Baseline Monitoring (Tracks 1 & 2):** Remote sensing analysis of the "Locked" Reference Area (Section 7) to determine the dynamic baseline rate ($\Delta C_{BLS,y}$) at verification.

15.3 | Digital MRV Mandate (Track 2 Safeguard)

15.3.1. | To ensure auditability for distributed smallholder activities and prevent "ghost acres," the activity developer shall utilize a mobile-based Digital MRV System.

- a. **Geospatial Traceability:** Every participating farm or tree cluster shall be registered with a unique ID and a GPS coordinates.
- b. **Evidence:** The system shall capture time-stamped, geotagged photographs of the planting sites to verify existence.
- c. **Audit Trail:** The system shall record the User ID, Date, and Time of every data entry to prevent retroactive fabrication.

15.4 | Data and parameters not monitored (Fixed *Ex-Ante*)

15.4.1. | The following parameters are determined ex-ante, are fixed for the crediting period, or are sourced from external, reputable databases where direct monitoring by the Activity Developer during each monitoring period is not required.

Parameter ID	1
Data/parameter:	R
Description	Root to shoot ratio (i.e., ratio of belowground (root) biomass to aboveground biomass, per unit area or per stem)
Data unit:	Dimensionless
Equations referred:	The parameter is listed in equation 11
Purpose of data	Activity removals; Calculation of Belowground Biomass (BGB)
Default Value(s)	Sourced based on specific species and ecological zone (e.g., 0.246 for applicable systems).
Source of data	Table 4.4 in Chapter 4 of the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Choice of data or measurement methods and procedures	Other sources: Calculated according to requirements of the methodology
Comments:	Fixed ex-ante based on the planted species and ecological zone.

Parameter ID	2
Data/parameter:	CF
Description	Carbon fraction of dry biomass
Data unit:	t C/t d.m
Equations referred:	
Purpose of data	Activity removals
Default Value(s)	0.47
Source of data	Other source: IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Choice of data or measurement methods and procedures	Other sources: IPCC
Comments:	Applies to aboveground and belowground woody biomass unless species-specific local data is available.

Parameter ID	3
Data/parameter:	ρ (Wood Density) and U_{model}
Description	Species-specific wood density required for allometric equations and associated model uncertainty
Data unit:	g/cm ³ or d.m./m ³ (Density) and Percentage (Uncertainty)
Equations referred:	Biomass allometry and Equation 25
Purpose of data	Activity removals Calculation of Activity Removals (Aboveground Biomass)
Default Value(s)	0.47
Source of data	Global Wood Density Database (Zanne et al., 2009), peer-reviewed scientific literature, or national forest inventories.
Choice of data or measurement methods and procedures	Other sources: IPCC
Comments:	Shall be species-specific. If the exact species is not listed, genus-level averages may be used conservatively.

Parameter ID	4
Data/parameter:	$EF_{gas}, EF_{CO2,i}, EF_N, GWP_{gas}$
Description	Emission Factors for biomass burning, fossil fuels, and synthetic nitrogen fertilizers, and Global Warming Potentials.
Data unit:	tCO ₂ e/unit mass or volume
Equations referred:	Equation 8,9,10,23,24,
Purpose of data	Activity emissions and Reversal emissions
Default Value(s)	Default
Source of data	Latest IPCC Guidelines and Assessment Reports as approved by Gold Standard for GWPs.
Choice of data or measurement methods and procedures	
Comments:	Fixed ex-ante but subject to updates at crediting period renewal based on GS4GG registry rules.

Parameter ID	5
Data/parameter:	DAF
Description	Downward Adjustment Factor
Data unit:	Fraction
Equations referred:	Equation 5 & 14
Purpose of data	Ambition adjustment to align baselines with Host Country Net-Zero trajectories.
Default Value(s)	0.0125 (1.25%)
Source of data	GS4GG Tool: Downward Adjustment Factor (DAF) Determination
Choice of data or measurement methods and procedures	
Comments:	Fixed Absolute Floor for the Land Use and Forestry (LUF) sector for the 2025 -2030 Application Period.

Parameter ID	6
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Data/parameter:	LF_{market}
Description	Market Leakage Factor
Data unit:	Fraction
Equations referred:	Equation 18
Purpose of data	Leakage emissions; Calculation of market leakage emissions.
Default Value(s)	0.00, 0.05, or 0.10.
Source of data	Methodology Section 10.3.
Choice of data or measurement methods and procedures	
Comments:	Assigned based on the type of baseline commodity displaced.

15.5 | Data and parameters monitored

15.5.1. | The following parameters shall be monitored during each monitoring period y .

Parameter ID	7	
Data/parameter:	A_i	
Description	Total eligible area of stratum i within the Activity Boundary	
Data unit:	ha	
Equations referred:	Equations 2,4	
Purpose of data	Baseline removals & Scaling of Activity removals; Calculation of Baseline Removals and scaling of Activity Removals	
Measurement methods and procedures	Combination of GIS coverage, ground survey data with GPS, and remote imagery (satellite or aerial photographs) used for delineation of the planting area.	
Entity/person responsible for the measurement	Field team deployed at the ground is responsible for the measurement	
Measuring instrument(s)	Type of instrument	NA
	Accuracy class	NA.
	Calibration requirements	NA
	Location	NA
Measurement intervals	Delineated at start date; verified at each monitoring period to reflect any changes.	

QA/QC procedures	Any imagery used shall be geo-registered referencing corner points or clear landmarks. Ground-truthed GPS points shall have a minimum accuracy of ± 5 meters.
Comments:	None

Parameter ID	8	
Data/parameter:	DBH and H	
Description	Diameter at Breast Height (DBH) and Tree Height (H)	
Data unit:	cm (DBH) and meters (H)	
Equations referred:	Used as independent variables in allometric equations for Activity Removals.	
Purpose of data	Activity removals; Calculation of Aboveground Biomass (AGB)	
Measurement methods and procedures	Field measurement: DBH directly measured in the field using DBH tapes or calipers. Height measured using a clinometer, hypsometer, or laser rangefinder on trees within PSPs or Representative Areas. Apply established best practices (Pancel, L., ed. 1993. Tropical forestry handbook. Berlin, Germany, Springer-Verlag. Volume 1, 738 pp.)	
Entity/person responsible for the measurement	Field team deployed at the ground is responsible for the measurement	
Measuring instrument(s)	Type of instrument	NA
	Accuracy class	NA.
	Calibration requirements	NA
	Location	NA
Measurement intervals	Measured at each verification event (Minimum every 5 years).	
QA/QC procedures	Annual calibration of measuring tapes and rangefinders. Independent internal audit re-measuring 5% of plots. Also Refer to QA/QC procedures prescribed under national forest inventory are applied. In the absence of these, QA/QC procedures from published handbooks, or from the IPCC GPG LULUCF 2003, are applied.	
Comments:	None	

Parameter ID 9	
Data/parameter:	<i>N</i>
Description	Total count of surviving planted trees (Population Size)

Data unit:	Integer (Count)	
Equations referred:	Census-based quantification approach (Section 9.4)	
Purpose of data	Activity removals; Calculation of Activity Removals (Track 2)	
Measurement methods and procedures	100% digital census conducted by smallholders or project staff, registered uniquely in the Digital MRV app with GPS coordinates and photographic evidence.	
Entity/person responsible for the measurement	Field team deployed at the ground is responsible for the measurement	
Measuring instrument(s)	Type of instrument	NA
	Accuracy class	NA.
	Calibration requirements	NA
	Location	NA
Measurement intervals	Annually monitored for survival; aggregated at each verification.	
QA/QC procedures	Digital logic algorithms audit system for duplicate coordinates or photos. Field auditor physically verifies a random 1% to 5% sample.	
Comments:	None	

Parameter ID	10	
Data/parameter:	$\Delta C_{BSL,i}$	
Description	Baseline Removal Rate (Observed natural regeneration)	
Data unit:	tC/ha/yr	
Equations referred:	Equation 2	
Purpose of data	Dynamic calculation of unadjusted Baseline Removals	
Measurement methods and procedures	Track 1: Remote Sensing analysis of the locked reference area. Track 2: Sourced from approved regional Dynamic Curves of NFI data.	
Entity/person responsible for the measurement	Field team deployed at the ground is responsible for the measurement	
Measuring instrument(s)	Type of instrument	NA
	Accuracy class	NA.
	Calibration requirements	NA

	Location	NA
Measurement intervals	Measured at each verification event	
QA/QC procedures	For Track 1, the VVB shall audit the Reference Area to ensure no anthropogenic land use change occurred.	
Comments:	None	

Parameter ID	11	
Data/parameter:	$M_{burn}, Vol_{fuel,i}, M_{N,i}$	
Description	Mass of dry biomass burned, volume of fossil fuel consumed, and mass of synthetic nitrogen fertilizer applied.	
Data unit:	tonnes d.m. , Liters/kg , tonnes N	
Equations referred:	Equations 8, 9, 10	
Purpose of data	Calculation of Activity Emissions (AE_y)	
Measurement methods and procedures	Estimated via field logs, GIS fire-scar mapping, fuel purchase receipts, and fertilizer application records. <i>(Note: Fossil fuel and fertilizer are exempt if using the 5% default deduction for Tracks 2/3).</i>	
Entity/person responsible for the measurement	Field team deployed at the ground is responsible for the measurement	
Measuring instrument(s)	Type of instrument	NA
	Accuracy class	NA.
	Calibration requirements	NA
	Location	NA
Measurement intervals	Recorded continuously as operations occur, aggregated at verification.	
QA/QC procedures	Cross-referenced with operational budgets, inventory logs, and satellite fire alerts (e.g., FIRMS).	
Comments:	None	

Parameter ID	12	
Data/parameter:	$Area_{def,y}, Y_{crop}$	
Description	Deforested area in the Leakage Belt (Track 1) and Crop Yields on project farms (Track 2).	

Data unit:	ha and tonnes/ha	
Equations referred:	Equations 16, 17	
Purpose of data	Quantification of Leakage Emissions (Activity Shifting)	
Measurement methods and procedures	Track 1: Remote sensing of the predefined leakage belt. Track 2: Farmer surveys or physical sampling of crop harvests.	
Entity/person responsible for the measurement	Field team deployed at the ground is responsible for the measurement	
Measuring instrument(s)	Type of instrument	NA
	Accuracy class	NA.
	Calibration requirements	NA
	Location	NA
Measurement intervals	At each verification event.	
QA/QC procedures	Remote sensing classification accuracy shall be >85%. Yield surveys conducted using strict randomized selection.	
Comments:	None	

15.6 | Sampling requirements

- 15.6.1. | The activity developer shall calculate the required sample size to achieve a target precision of $\pm 10\%$ of the mean at the 90% Confidence Interval (CI). If this precision is not met, the Uncertainty Deduction defined in Section 14.3 shall be applied.
- 15.6.2. | **Method A: Stratified Random Sampling (Track 1):** Applicable to Module 1 (Commercial Forestry), Module 2 (Restorative Forestry) or Module 3 (Production Agroforestry) and for Sub-Module A (revegetation).
- Stratification:** The activity boundary shall be stratified into homogeneous units based on species, age cohort, and biophysical conditions (e.g., soil type, slope).
 - Permanent Sample Plots (PSPs):** The activity developer shall establish a network of PSPs using systematic sampling with a random start within a stratum.
 - Allocation:** The total number of permanent sample plots required to achieve the target precision shall be calculated based on the variance of biomass or carbon stock estimates. The number of plots (n) per stratum shall be calculated using the *Neyman's optimal allocation* method to

minimize variance (and therefore the standard error) of the stratified mean estimate for a fixed total sample size.

- a. **Location:** Plot centres shall be randomly generated via GIS and physically or digitally marked.
- b. **Size:** Sufficient to capture at least 10–20 trees per plot (typically 0.04 to 0.1 ha).
- c. **Measurement:** All trees within the plot boundary (Minimum DBH that selected allometric equations applicable) shall be measured for DBH and Height.

15.6.2.1 | Under Neyman⁴ allocation, the number of plots allocated to stratum h is calculated as:

$$n_h = n \times \frac{A_h \cdot \sigma_h}{\sum_{i=1}^L (A_i \cdot \sigma_i)} \quad (\text{Eq-28})$$

Where

Parameter	Description	Units
n_h	Number of plots allowed in stratum h	Dimensionless (Number)
n	Total sample size (total number of plots across the activity area)	Dimensionless (Number)
A_h	Area of stratum h	ha
σ_h	Standard deviation of biomass or carbon stock in stratum h	tC/ha or tCO ₂ e/ha
L	Total number of strata	Dimensionless (Number)
$\sum_{i=1}^L (A_i \cdot \sigma_i)$	The sum of the product of Area and Standard Deviation across all strata (from 1 to L)	tC or tCO ₂ e

15.6.3. | **Method B: Census-Based / Double-Phase Sampling (Track 2):** Applicable to Module 4 (Subsistence Agroforestry), Small scale or Micro scale Module 3 (Production Forestry), Module 4 (Revegetation) or for Sub-Module A (Smallholder). For dispersed smallholder systems, a pure plot-based approach yields excessive variance. A hybrid **Census-Sample** approach is required:

⁴ Neyman allocation minimizes the variance of the stratified mean estimator. As the standard error is the square root of this variance, minimizing variance also minimizes the standard error of the estimate. Accordingly, Neyman allocation is fully consistent with achieving the target precision requirement expressed at the 90% CI. Within-stratum standard deviations (σ_i), derived from pilot sampling, previous monitoring events, or comparable reference data, shall be used for allocation. Where only within-stratum standard errors are available, they shall be converted to standard deviations prior to application of Neyman allocation, whereby the Neyman allocation formula itself remains unchanged.

- a. **The Census (N):** The Total Tree Count (N) is derived from the **100% Digital Census** recorded in the MRV system. As a full population count, this parameter has zero sampling error (subject to QC audits).
- b. **The "Mean Tree" Sample:** To estimate the biomass of the "Average Tree" (C_{sample}), the activity developer shall sample a Representative Cohort.
 - i. **Sampling Frame:** The activity developer may utilize the representative area from lands / farms within the project area selected for Reversal Monitoring (Section 13.4) as the biomass sample frame.
 - ii. **Sample Size:** A minimum of **30 trees** per species/age cohort shall be measured to establish the allometric mean.
 - iii. Calculation:

$$Total\ biomass = N_{census} \times C_{sample} \quad (Eq-29)$$

Where

Parameter	Description	Units
N_{census}	Total Tree Count from the 100% digital census	Dimensionless (Number)
C_{sample}	Biomass of the average tree	tC of a tree

15.6.4. | **Method C: Allometric / Destructive Sampling (Track 3):** Applicable to Module 4 (Revegetation) and applicable configurations of Sub-Module A.

- a. **Destructive Sampling:** Where published allometric equations are unavailable for the specific non-tree species (e.g., bamboo, shrubs), the Developer shall harvest biomass from randomly selected quadrats (e.g., 1m × 1m) to develop a local biomass density factor.
- b. Harvested wet weight shall be converted to dry matter via laboratory oven-drying of subsamples until constant weight is achieved.

15.7 | QA/QC and Data Management

15.7.1. | The activity developer shall implement Quality Assurance (QA) and Quality Control (QC) procedures to ensure data integrity.

15.7.2. | **Data Archiving and Custody**

- a. **Central Database:** All monitoring data (field sheets, digital logs, GPS files, remote sensing imagery) shall be centralized in a secure, redundant cloud database.
- b. **Retention:** Data shall be retained for the duration of the Crediting Period plus 2 years.

- c. **"Locked Map" Archive:** The specific shapefiles constituting the Dynamic Reference Area (Section 7) shall be archivally preserved (cryptographically hashed) at the time of Validation. At each Verification, the file hash shall be checked to ensure pixel coordinates were not altered to manipulate the baseline.
- 15.7.3. | **Field Measurement QC (Internal Audit):** Prior to external verification, the Activity Developer shall conduct an Internal Audit.
- a. A supervisor or independent auditor shall re-measure a random sub-sample of **5%** of the Sample Plots (Track 1) or Representative Area (Track 2).
 - b. **Pass Criteria:** The deviation between the Original Measurement and the Audit Measurement shall be $\leq 5\%$.
 - c. **Action:** If the deviation exceeds 5%, the entire batch of data from that field team shall be rejected and re-measured.
- 15.7.4. | **Digital Logic Checks (Track 2 Safeguard):** The Digital MRV system shall apply automated logic checks:
- a. **Geofencing:** The system shall prevent registration of trees outside the eligible Activity Boundary.
 - b. **Duplicate Detection:** Automated algorithms shall flag duplicate GPS coordinates or duplicate photographs used across multiple farm records.
 - c. **Growth Limits:** The system shall flag tree growth rates that exceed biological maximums (e.g., >5m height growth in Year 1) for manual review.
- 15.7.5. | **Remote Sensing QA/QC:** For Baseline Monitoring (Tracks 1 & 2):
- a. **Accuracy Assessment:** Land Use Change (LUC) maps derived from satellite imagery shall achieve an Overall Accuracy of > 85%, verified using high-resolution ground truth or aerial imagery (Olofsson et al., 2014).
 - b. **Cloud Cover:** Optical imagery used for the "Locked Map" analysis shall have < 20% cloud cover over the Reference Area to ensure data viability.

16 | APPLICATION TO PROGRAMME OF ACTIVITIES

16.1 | General Requirements

- 16.1.1. | The activity developer (acting as the Coordinating/Managing Entity - CME) may apply this methodology within a Programme of Activities (PoA). The PoA framework allows for the replicable inclusion of an unlimited number of Voluntary Project Activities (VPAs) under a single administrative umbrella.
- 16.1.2. | The PoA shall strictly adhere to the *GS4GG Programme of Activity Requirements*, and the specific procedures defined below.

16.2 | PoA Design and Boundary

16.2.1. | The CME shall define the programmatic framework in the PoA Design Document (PoA-DD).

- a. **PoA Spatial Boundary:** The PoA-DD shall define the physical geographical boundary (e.g., National, Sub-national, or defined Eco-regions) within which all VPAs shall be located.
- b. **VPA Types (Homogeneity):** To accommodate the Modular Architecture of this methodology, the PoA-DD shall define distinct VPA Types.
 - a. **Constraint:** A single VPA Type shall correspond to a single Compliance Track (Section 3) and Activity Module. VPAs shall not mix Tracks. Example Structure:
 - i. VPA Type 1: Large Scale Restorative Forestry (Module 2, Track 1).
 - ii. VPA Type 2: Smallholder integrated subsistence (Module 4, Track 2).
- c. **Real Case VPA:** At the time of Design Certification, the CME shall submit a complete VPA Design Document (VPA-DD) for at least one "Real Case" VPA for each VPA Type included in the PoA. This sets the verified precedent for all subsequent inclusions.

16.3 | Criteria for Inclusion of VPAs

16.3.1. | The PoA-DD shall establish objective Inclusion Criteria that every new VPA shall meet. The CME is responsible for conducting an "Inclusion Check" for every new VPA prior to its formal inclusion.

16.3.2. | At a minimum, the Inclusion Criteria shall confirm:

a. General Eligibility

- i. **Location:** The VPA is located within the defined PoA Boundary.
 - ii. **Land Eligibility:** The VPA meets the Applicability Conditions of Section 3 (e.g., Non-Forest Status >10 years, Wetland Exclusion).
 - iii. **Track Compliance:** The VPA adheres to the monitoring method (e.g., Census vs. Plot) and baseline method (e.g., Regional vs. Matched) assigned to its VPA Type.
 - iv. **Digital Onboarding (Track 2):** For Smallholder VPAs, the CME confirms that every participating land parcel has been registered in the Digital MRV System with a unique ID and valid geolocation.
- b. **Additionality (Programmatic Level):** In accordance with *GS4GG PoA Requirements*, additionality is demonstrated at the PoA level to reduce transaction costs.

- i. **Deemed Additionality:** If the VPA Type falls under the **Positive List** (Section 6.2) (e.g., Smallholder/LDC), the VPA is deemed additional upon meeting the Inclusion Criteria.
 - ii. **Unit-Level Check:** For VPA Types not covered by the Positive List (e.g., Commercial Track 1), the Inclusion Criteria shall require a VPA-specific **Investment Analysis** (Section 6.6) or confirmation that the VPA's parameters fall within the non-viable range of a "Matrix Screening" approved in the PoA-DD.
- c. Baseline Assignment (Inheritance):** To simplify baseline setting, VPAs may "inherit" parameters defined in the PoA-DD:
- i. **Reference Area Inheritance (Track 1):** A new VPA may utilize a "Master Reference Area" established in the PoA-DD *only if* the CME demonstrates via the **Covariate Balance Test** ([Annex 1](#)) that the new VPA is biophysically equivalent ($SMD < 0.1$) to the Master Reference Area.
 - ii. **Regional Inheritance (Tracks 2 & 3):** The PoA-DD defines valid Regional Dynamic Curves (Section 7.3.2) for specific Eco-Zones. A new VPA located in "Zone A" automatically applies "Curve A."

16.4 | Monitoring and Sampling in PoAs

16.4.1. | In accordance with the [GS4GG Methodology for Sampling and Estimation of Parameters](#), the CME may elect to conduct monitoring using a Cross-VPA Sampling approach.

- a. **Homogeneity Condition:** Cross-VPA sampling is permitted only for VPAs belonging to the same VPA Type (same Track, species mix, and age cohort). Track 1 and Track 2 VPAs shall never be pooled into the same sample.
- b. **Sample Size:** The sample size (n) is calculated for the aggregated area of the grouped VPAs to achieve $\pm 10\%$ precision at the 90% Confidence Interval.
- c. **Census Requirement (Track 2):** For Track 2, the 100% Digital Census (N) is mandatory for every farm in every VPA. Sampling is permitted only for the determination of the biomass parameters (e.g., Mean Tree Size).
- d. **Verification:** The VVB shall verify the CME's management system and conduct site visits to a random sample of VPAs and a random sample of plots/farms within those VPAs.

16.5 | CME Liability

16.5.1. | The CME assumes full liability for the validity of included VPAs. If a Verification Audit reveals that a VPA (or a land parcel within a VPA) was included in error

(e.g., land was already forest), the CME shall compensate for any excess credits issued by retiring an equivalent amount from their own account.

17| RENEWAL OF CREDITING PERIOD

17.1 | General Requirements

- 17.1.1. | In accordance with the [GS4GG Land Use & Forests Activity](#) Requirements, the Crediting Period for ARR activities is typically 40 to 100 years. However, to ensure the activity remains aligned with evolving science, regulatory conditions, and national climate ambition, the crediting status shall be renewed every 5 Years (the Renewal Cycle).
- 17.1.2. | The requirements in this section apply to:
- a. **Standalone Projects:** The renewal of the specific project activity.
 - b. **Programme of Activities (PoA):** The renewal of individual Voluntary Project Activities (VPAs) based on their specific start dates.
- 17.1.3. | The activity developer shall successfully complete a **Renewal Audit** to update the baseline parameters (if applicable) and reconfirm eligibility for the subsequent 5-year period.

17.2 | Reassessment of the Baseline Scenario

- 17.2.1. | To prevent the "locking in" of obsolete data in a rapidly changing environment, the methodology mandates specific updates to the baseline parameters. The scope of this update depends on whether the renewal coincides with the expiration of the **Baseline Validity Period (10 Years)**. For individual VPAs under a PoA, the 10-year Baseline Validity Period shall be tracked and renewed independently based on the specific start date of each individual VPA, not the overarching PoA start date.
- 17.2.2. | **Redetermination of BAU:** The Business-As-Usual (BAU) scenario and the quantification of BAU emissions and/or removals shall be formally redetermined at each crediting period renewal.
- 17.2.3. | **Reference Area Audit:** The "Locked Map" (Dynamic Reference Area) established in Section 7 has a maximum validity of 10 Years. Therefore, the technical requirements at renewal differ based on the cycle:
- 17.2.3.1 | **At the 5-Year Renewal (Interim Update)**
- a. **Track 1:** The Spatial Boundary of the Reference Area remains Locked. The activity developer continues monitoring the same control pixels established at validation. No new matching is required.
 - b. **Track 2:** The activity developer shall adopt the most recently published Regional Dynamic Curve or National Forest Inventory (NFI) dataset for the relevant Eco-Zone.

- c. **Track 3:** The linear trend projection is maintained unless a significant divergence in regional climate trends is observed (e.g., sudden greening due to increased rainfall).

17.2.3.2 | At the 10-Year Renewal (Full Reset)

- a. **Unlock:** The spatial definition of the Reference Area (for Track 1) is opened for audit.
- b. **Disturbance Audit:** The Verification Body (VVB) shall verify if any reference pixels were subject to Anthropogenic Land Use Change (e.g., urbanization, conversion to intensive cropland, or infrastructure development) during the previous decade.
- c. **Replacement:** Compromised pixels (those no longer representing the "Natural Regeneration" counterfactual) shall be removed and replaced with new matches from the Donor Pool using the matching procedure in Annex 1.
- d. **Re-Lock:** The updated map is "Locked" for the subsequent 10-year period.

17.3 | Update of Downward Adjustment Factor (DAF)

17.3.1. | To ensure continuous alignment with the goals of the Paris Agreement, the Downward Adjustment Factor (DAF) is subject to periodic updates. In accordance with Section 6.1 of the GS4GG Tool: Downward Adjustment Factor Determination (PAA MT400-05), DAF values are fixed for specific calendar Application Periods (e.g., 2026-2030, 2031-2035).

- a. **Application Calendar Independence:** The application of the DAF values is determined by the calendar year in which the removals occur. This application is strictly independent of the start date, duration, or renewal schedule of a specific project activity's crediting period.
- b. **Automatic Transition:** Therefore, during a crediting period, the activity shall automatically transition to apply the updated default DAF values (or the Absolute Floor, if applicable) published by Gold Standard corresponding to the calendar year in which the vintage occurs.

17.4 | Reconfirmation of Additionality

17.4.1. | The activity developer shall demonstrate that the activity remains additional under the current regulatory and economic conditions.

17.4.2. | **Regulatory Surplus:** The activity developer shall demonstrate that no new laws, regulations, statutes, or binding mandates have been enacted since the last validation that explicitly require the maintenance of the trees established by the project.

- a. **Adjustment:** If a new law mandates a minimum tree cover (e.g., 10% on agricultural land), the **Crediting Baseline Floor** (Section 7.5 and 7.9) shall be adjusted upward to reflect this mandatory compliance

level, ensuring the project is only credited for voluntary action above the law.

17.4.3. | **Financial Need (OFN)**

- a. **Applicability:** Mandatory for Track 1 (Commercial) activities.
- b. **Requirement:** The Activity Developer shall provide an updated financial analysis or qualitative narrative demonstrating that carbon revenues are still required to sustain the activity (e.g., covering the costs of guarding, fire management, or replanting).
- c. **Exemption:** Track 2 (Smallholder) and Track 3 (Restoration) activities are exempt from the OFN demonstration, provided they remain non-commercial (no timber harvesting) and meet the eligibility criteria of the Positive List.

ANNEX 01: TECHNICAL PROCEDURES FOR BASELINE DETERMINATION

Related Section: Section 7 (Baseline Scenario) **Applicability:** Mandatory for all Compliance Tracks.

A1. | Governance:

To resolve the issue of "shifting baselines" and prevent *ex-post* manipulation (e.g., redrawing boundaries to exclude regenerating areas), this methodology separates the Spatial Domain (where we measure) from the Temporal Data (what we measure).

A1.1 Overview: The geographic boundaries of the Reference Area (RA) shall be fixed ("*Locked*") prior to the analysis of carbon stock data for any given monitoring period.

A1.2 Ex-ante:

- a. **Timing:** The Reference Area boundaries are defined and validated at Design Certification (Year 0).
- b. **Validity:** The map remains locked for the Baseline Validity Period (10 years).
- c. **The "Blind" Check:** During annual monitoring, the activity developer extracts the mean Carbon Stock Change ($\Delta C_{ref,y}$) from the exact pixel coordinates defined in the Locked Map.

A1.3 Ex-Post Audit

At every verification event, the VVB shall conduct an audit of the Locked Reference Area.

- **Trigger:** Pixels within the RA that undergo abrupt Anthropogenic Land Use Change unrelated to natural regeneration shall be flagged. This includes:
 - Conversion to Settlement/Infrastructure (Urbanization).
 - Conversion to Intensive Agriculture (Plowing).
 - New Mining or Road Construction.
- **Evidence:** Change shall be confirmed via visual inspection of high-resolution imagery (<5m) or official land registry data.
- **Action:**
 - **Scenario A ($\leq 10\%$ affected):** Affected land parcels are masked out for the current monitoring period. Practically, this means the affected pixels shall be removed from the calculation, and the mean baseline rate shall be recalculated using only the remaining valid pixels.
 - **Scenario B ($> 10\%$ affected):** The Reference Area is deemed "Compromised." The Activity Developer shall initiate a **Re-Matching Event** (see Section 17.2) to replace the compromised pixels with new matches from the Donor Pool.

User Guideline: Why Lock the Map? Locking the map at the start ensures the baseline is a true, unbiased "control group," similar to a clinical drug trial.

A 2 | Track 1 Technical Protocol: Spatially Explicit Matched Dynamic Baseline (SEMDB)

- **Applicability:** Large Scale Projects.
- **Purpose:** To construct a "Synthetic Control" group that isolates the project's impact from environmental forcing.

A.1 Data Sources: The Activity Developer shall use the following (or equivalent high-resolution) data sources for covariate extraction. To ensure practical application, the use of cloud-based geospatial processing platforms (e.g., Google Earth Engine) is highly recommended to handle the raster data volume:

Covariate Type	Recommended Dataset	Resolution	Reference
Biomass, NDVI or other biomass proxy	Landsat (Hansen/GFC) or Sentinel-2	30m / 10m	Hansen et al. (2013)
Elevation/Slope	SRTM (Shuttle Radar Topography Mission)	30m	Farr et al. (2007)
Precipitation	CHIRPS or WorldClim (BioClim)	~5km / 1km	Fick & Hijmans (2017)
Soil Carbon/Texture	ISRIC SoilGrids 2.0	250m	Poggio et al. (2021)
Accessibility	OpenStreetMap (Roads/Settlements)	Vector	OpenStreetMap Contributors

A.2.2 Matching Procedure (Step-by-Step)

Step A: Define the Donor Pool:

Identify all pixels within the Level-3 Ecoregion (Olson et al., 2001) that:

1. Are classified as **Non-Forest** at Year 0.
2. Are **> 5 km** from the Activity Boundary (to prevent Leakage contamination).

Step B: Covariate Matching Algorithm

The developer shall use Mahalanobis Distance Matching (MDM) or Propensity Score Matching (PSM) to pair each Project Pixel with the most similar pixel(s) from the Donor Pool.

- **Algorithm:** Nearest Neighbour Matching (1-to-1 or 1-to-k with replacement).
- **Distance Metric (D_{ij}):** If using Mahalanobis Distance, it is calculated as:

$$D_{ij} = \sqrt{(X_i - X_j)^T \Sigma^{-1} (X_i - X_j)} \quad (\text{Eq-30})$$

Where X_i and X_j are vectors of covariates for project unit i and control unit j , and Σ^{-1} is the inverse covariance matrix.

Where

Parameter	Description	Units
D_{ij}	Mahalanobis Distance between project pixel i and control pixel j	Dimensionless
X_i	Vector of covariates for project unit i	Vector array
X_j	Vector of covariates for control unit j	Vector array
Σ^{-1}	Inverse of the covariate covariance matrix	Matrix

Step C: Mandatory Covariates & Tolerances

Matches are valid only if they satisfy the Matching Tolerances in Table A3.

Table A.3 Mandatory Matching Covariates

Category	Covariate	Matching Tolerance (Calliper)
Edaphic	Soil Organic Carbon (SOC)	$\pm 10\%$ of mean in the project area
	Soil Texture (Clay/Sand)	Exact Class Match
Pre-project biomass	NDVI or other biomass proxy	$\pm 10\%$ of Mean in the project area
Topography	Elevation	± 200 meters
	Slope	± 10 degrees
Climatic	Precipitation (MAP)	Same Isohyet / Ecoregion
History	Vegetation Trend (5-yr)	$p > 0.05$ (Parallel Trends)
Access	Distance to Roads	± 1 km
Tenure	Legal Status	Exact Status Match (e.g., Private vs. Private)

Step D: Statistical Validation (The "Twin Test")

To demonstrate validity, the Developer shall report the Standardized Mean Difference (SMD) for all covariates.

$$SMD = \frac{\bar{X}_{project} - \bar{X}_{ref}}{\sqrt{\frac{S_{project}^2 + S_{ref}^2}{2}}} \quad (\text{Eq-31})$$

Where

Parameter	Description	Units
SMD	Standardized Mean Difference	Dimensionless

$\bar{X}_{project}$	Mean value of the covariate in the Project Area	Variable
$\bar{X}_{ref}$	Mean value of the covariate in the Reference Area	Variable
S^2	Variance of the covariate in the respective area	Variable squared

- **Pass Criteria:** $SMD < 0.1$ for all key covariates (Austin, 2011).
- **Fail Criteria:** If $SMD > 0.1$ for any covariate, the match is deemed potentially biased. The Donor Pool shall be expanded, or the matching algorithm refined.

Step E: Parallel Trends Test

Demonstrate that the Project and Reference areas followed similar trajectories prior to intervention.

- **Test:** Plot the mean NDVI of both groups for Years -5 to -1.
- **Requirement:** The trend lines shall not diverge significantly (Visual inspection or Placebo Test with $p > 0.05$) (Abadie, 2005).

User Guideline:

Matching Software Do not simply pick a nearby field as a reference. You shall run a statistical script to scan the entire region. Recommended open-source packages include:

- *R: MatchIt, optmatch*
- *Python: CausalInference, PyMatch*

A 3 | Track 2 Technical Protocol: Regional Aggregation

Applicability: Small and micro-scale projects.

Purpose: To simplify baselines using robust regional averages, lowering barriers to entry.

1. Stratification Rules

The Reference Area shall be stratified to match the Project Area's Land Class.

- Source: FAO Land Cover Classification System (LCCS), ESA WorldCover, or national equivalent. If the project is restoring "Degraded Shrubland," the Reference Rate shall be derived strictly from other "Degraded Shrubland" land parcels. Mixing in "Bare Soil" or "Forest" pixels will skew the baseline.

2. Data Hierarchy

The Developer shall select the baseline rate ($\Delta C_{regional}$) using the following hierarchy:

Tier 1: Approved Regional Curves

- Source: Dynamic curves published by the Gold Standard Secretariat or Host Country Government for specific Eco-Zones.
- Usage: Mandatory if available.

Tier 2: Aggregated National Forest Inventory (NFI)

- Source: Aggregated data from official NFI plots.
- Constraint: Shall include ≥ 30 plots within the same administrative Level-1 district.
- Constraint: Data collection date shall be < 5 years from project start.

Tier 3: Simplified Remote Sensing

- Source: A random sample of at least 100 points spatially distributed across the Reference Stratum, analysed for biomass trends using open-source tools (e.g., temporal segmentation of Landsat series using LandTrendr or BFAST algorithms). This tier is applied with necessary ground truthing or a-priori information.
- Conversion: Observed trends in mean annual NDVI shall be converted to biomass using a locally calibrated allometric factor derived from peer-reviewed literature or direct physical ground-truthing. Generic uncalibrated conversions are prohibited.

A 4 | Track 3 Technical Protocol: Static Baseline

Applicability: Modality 4 (Non-Tree / Revegetation).

Purpose: To provide a conservative, predictable baseline for non-woody systems where optical noise is high.

1. Trend Analysis

Calculate the linear regression of the Vegetation Index (VI, e.g., NDVI) over the historical reference period (10 years).

$$VI_y = \alpha + \beta \cdot Year + \varepsilon \quad (\text{Eq-32})$$

Where

Parameter	Description	Units
VI_y	Mean Vegetation Index value for year y	Index Value
α	The intercept of the linear trend line	Index Value
β	The slope (annual rate of change) of the trend line	Index/ yr
ε	Random error term	Dimensionless

2. Forward Projection & Capping

- If $\beta \leq 0$ (Negative/Flat Trend): Baseline Removals = 0.
- If $\beta > 0$ (Positive Trend): Project the trend forward linearly.
- The Shrub Cap: The projected baseline biomass shall not exceed the theoretical carrying capacity (B_{max}) of the ecosystem.

$$BL_{unadj,y} = \text{Min}(\text{Project trend}, B_{max})$$

- **Default B_{max} :** Unless measured locally, use IPCC Tier 1 defaults for "Mature Shrubland" in the relevant climate zone (IPCC, 2019).

A 5 | Hybrid Monitoring

Applicability: Projects using remote sensing for baseline assessment.

Problem: Optical satellites cannot reliably detect seedlings (<1.5m). Comparing "Project Bare Soil" to "Reference Bare Soil" in early years yields noise, not signal (Olofsson et al., 2014).

1. Phase 1: Establishment (Years 0–3)

- **Baseline Assumption:** $BL_{unadj,y} = 0$
- **Crediting:** Credits are based solely on the accumulation of biomass in the planted trees (measured via Census or Stratified Plots).
- **Disturbance Spike Check:** During Phase 1, the Reference Area is monitored for sudden "Spikes" (e.g., NDVI increase > 0.2 in a single year). A spike indicates rapid recovery from a disturbance (e.g., fire/harvest) that happened *just before* the project start. If a spike is detected, the Zero assumption is void, and the spike magnitude is deducted.

2. Phase 2: Canopy Activation (Year 4+)

- **Trigger:** The transition to Phase 2 occurs when the Project Area achieves a mean Canopy Closure of >10% OR average tree height >1.5 meters.
- **Action:** The full SEMDB (Track 1) or Regional Curve (Track 2) comparison activates.

A 6 | Uncertainty Quantification (UNC_{BSL})

The statistical uncertainty of the baseline matching shall be calculated at the 90% Confidence Interval (CI) and applied as an independent conservativeness deduction (UNC_{BSL}) in Equation 5 (Section 7.9).

1. Track 1 Calculation For the SEMDB approach, the uncertainty is the Standard Error (SE) of the mean carbon stock change of the matched control pixels, expressed as a fractional percentage of the mean change.

$$UNC_{BSL} = \frac{1.645 \times SE_{control}}{\Delta C_{control}} \quad (\text{Eq-33})$$

Where:

$$SE_{control} = \frac{\sigma_{control}}{\sqrt{N_{control}}} \quad (\text{Eq-34})$$

Where

Parameter	Description	Units
$\sigma_{control}$	Standard Deviation of ΔC in the Reference Area pixels.	
$N_{control}$	Number of pixels in the Reference Area.	Dimensionless

2. Track 2 & 3 Defaults

Due to the lack of explicit spatial pixel matching, higher default uncertainty deductions apply as established in Section 7.4:

- **Track 2 Default:** 10% (0.10)
- **Track 3 Default:** 20% (0.20) (Reflecting the high variance in non-tree biomass estimation).

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